

Review

Transfer learning and its extensive appositeness in human activity recognition: A survey

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ABSTRACT

In this competitive world, the supervision and monitoring of human resources are primary and necessary tasks to drive context-aware applications. Advancement in sensor and computational technology has cleared the path for automatic human activity recognition (HAR). First, machine learning and later deep learning play a cardinal role in this automation process. Classical machine learning approaches follow the hypothesis that the training, validation, and testing data belong to the same domain, where data distribution characteristics and the input feature space are alike. However, during real-time HAR, the above hypothesis does not always true. Transfer learning helps in an extended manner to transfer the required knowledge among heterogeneous data of various activities. To display the hierarchical advancements in transfer learning-enhanced HAR, we have shortlisted the 150 most influential works and articles from 2014–2021 based on their contribution, citation score, and year of publication. These selected articles are collected from IEEE Xplore, Web of Science, and Google Scholar digital libraries. We have also analyzed the statistical research interest related to this topic to substantiate the significance of our survey. We have found a significant growth of 10% in research publications related to this domain every year. Our survey provides a unique classification model to delineate the diversity in transfer learning-based HAR. This survey delves into the world of HAR datasets, exploring their types, specifications, advantages, and limitations. We also examine the steps involved in HAR, including the various transfer learning techniques and performance metrics, as well as the computational complexity associated with these methods. Additionally, we identify the challenges and gaps in HAR related to transfer learning and provide insights into future directions for researchers in this field. Based on the survey findings, researchers prefer the inductive transfer method, feature learning transfer mode, and cross-action transfer domain more over others due to their superior performance, with respective popularity scores of 55%, 40.8%, and 50.2%. This review aims to equip readers with a comprehensive understanding of HAR and transfer learning mechanisms, while also highlighting areas that require further research.

1. Introduction

In this “Population Explosion” era, humans have become a great resource while furnishing an intelligent task. Human Activity Recognition (HAR) can be potentially used in diverse applications like healthcare: to monitor the physical activities of patients, such as the elderly or those with chronic diseases, to assess their functional status and detect changes in their health, sports and fitness: to track and monitor the physical activities of athletes and fitness enthusiasts to improve their training and performance that help optimize their training regimens and prevent injuries, home automation: to control and automate devices and appliances in a home based on the detected activity of the occupants, security: to detect and alert security personnel to unusual or suspicious activities (Aslam & Kolekar, 2022; Aslam, Rai, & Kolekar,

2022), gaming and virtual reality: to enhance gaming experiences and virtual reality simulations by enabling more natural and intuitive interactions with the virtual environment, transportation: to monitor and analyze the driving behavior of individuals, such as speed, braking, and acceleration, to assess their driving habits and identify areas for improvement which reduce accidents and improve road safety, workplace safety: to monitor and detect unsafe work practices and behaviors in hazardous work environments which help prevent accidents and injuries and improve workplace safety, and personalized recommendations: to personalize recommendations and services based on a user’s activity patterns and preferences. In this machinery world, every organization still needs human resources for proper functioning. Therefore,

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every sector needs to take account human activities at utmost priority. Apart from that, the importance of HAR cannot be forgotten in building an intelligent society. However, manual bookkeeping of such activities can be a tidy and faulty human practice. So, researchers have taken advantage of different machine learning and deep learning techniques to automate these recognition processes. Any unusual and suspicious activity will ring the bell for either auto-correction or manual intervention. Nowadays, most functionaries deploy automatic recognition systems to make this task cost-effective and error-free.

Classical machine learning techniques get their success while working in a static environment. It runs through the hypothesis that the training, validation, and testing data belong to the same domain, where data distribution characteristics and the input feature space are alike. However, the scenario varies regularly, including working patterns, working environments, and the working person itself. Due to the mutable environment, the significance of older training data is inadequate for the current recognition task. This problem puts an extra burden on collecting and labeling a new set of training data. The “knowledge transfer” is a suitable option for the researcher to eliminate similar domain and task assumptions. Through the help of transfer learning (TL), we can easily exploit the older activity samples and utilize the valuable information to enhance the model performance.

The complexity in real-world scenarios always fluctuates from one environment to another, from one device to another, and from one setting to another. The researcher has pushed many efforts to get rid of these hurdles, limitations, and boundaries. This continual energy that has been put in this direction came into the picture recently when automated systems replace and ease the hard work behind human intervention in different real-world problems. Nowadays, the universality of machine learning and deep learning has become sky-high. The deep learning researchers initially encountered many difficulties and glitches while solving real-world data problems. Those problems had put future machine learning in jeopardy. Collection and pre-processing of training data are one of them (Chen & Lin, 2014). Deep learning algorithms are always data-hungry and behave well while executing large amounts of data. Cost and effort in labeling, extinction, and unavailability of real-world data compelled the research society to think about “knowledge transfer”. We can re-utilize the obsolete labeled data and decrease the labeling cost while solving infrequent data problems. This knowledge transfer is popularly referred to as transfer learning (TL). It accomplished its purpose in solving complicated heterogeneous tasks. Through the help of TL, we can easily exploit the older samples and utilize the valuable information to enhance the accuracy of a model. TL has attracted massive attention since 1997 within machine learning society and now in deep learning. Video classification (Karpathy et al., 2014), image classification (Han, Liu, & Fan, 2018; Ray & Kolekar, 2022), text classification (Moriya & Shibata, 2018), sentiment classification (Fei & Li, 2020), human activity recognition (Ray, Kolekar, Balasubramanian, & Hafiane, 2023), pose estimation (Topham, Khan, Al-Jumeily, & Hussain, 2022), and Wi-Fi signal-based location estimation (Lu et al., 2016) are among the most significant areas where practices of different transfer learning techniques are highly beneficial.

The style of performing any human activity has changed over the period. The equipment needed to accomplish an activity, its environmental state, and the sensor specification has also been updated regularly. These above factors are responsible for the mutation of datasets linked to that activity. Apart from this, enormous effort has been put into labeling these activities. To get rid of these challenges and hardships, transfer learning comes into the picture in the HAR domain. The extensive use of transfer learning techniques in HAR has multiple benefits that compelled researchers to design various deep transfer learning models. Many surveys have been reported to keep records of these designs along with their background settings, contributions, performances, complexities, benefits, and limitations. Most of them restrict their reviews to wearable, or ambient, or vision-based sensor backgrounds. Nweke, Teh, Al-Garadi, and Alo (2018) investigated deep

learning-based models that were applied to various activity datasets collected by mobile and wearable devices. Wang, Chen, Hao, Peng, and Hu (2019) analyses deep learning techniques based on the sensor modality. This paper investigates the transfer learning-based HAR by considering all sensing aspects (wearable, non-wearable and ambient) and learning aspects (probabilistic learning, machine learning, and deep learning), rather than limiting it to a constrained sensing and learning domain. Many studies have been done on transfer learning and activity recognition separately. To the best of our knowledge, a few of them investigated the activity recognition based on the transfer learning technique, and the number becomes scarce in our particular research domain. The contribution of our paper can be summarized as given below.

1. This paper provides a unified review on HAR using TL by summarizing 150 significant research articles (from 2014 to 2021) explored from IEEE, Elsevier, and Google Scholar platforms. To the best of our knowledge, this (Cook, Feuz, & Krishnan, 2013) is the last published review article related to our specific research domain comprising research papers up to 2013.
2. We analyze the statistical research interest in transfer learning-based HAR by following the Okali and Schabram methodology (Okoli & Schabram, 2010). This paper investigates the information regarding the most followed state-of-the-art algorithms and datasets related to our topic of interest.
3. To the best of our knowledge, we propose a three-modular format for TL classification, considering the representation of data, analysis of data, and the relation between data and environment as three referencing factors.
4. We investigate the research priority related to type, mode, domain, and label availability of transfer learning, as shown in Fig. 1. This paper also inspects the relative performance and computational complexity of the different transfer learning-enhanced recognition models.

The rest of the paper is organized as follows. Section 2 discusses the key specifications of this paper by comparing the related articles from TL-based HAR backgrounds. Section 3 introduces various HAR datasets, their classification, and hierarchical tabular representation with the specification. Section 4 introduces a detailed study of various steps involved in HAR. We categorize all methods that are used to implement these steps. In Section 5, we classify the transfer learning techniques concerned with HAR into a three-way representation format. Here, we have labeled related articles based on the model architecture, mode of transfer, area of application, and source/target datasets. Section 6 summarizes and discusses transfer learning enhanced HAR articles from 2014 to 2021 in terms of performance (accuracy), transfer mode, target dataset, and mode of application. Computational complexity of various related articles is also discussed in this section. Section 7 reports different challenges and future scope concerned to our theme. Finally, Section 5 concludes the paper by featuring its high points.

2. Literature survey

The theory of transfer learning has been continuously investigated and exploited due to its unparallel contribution to boosting performances in various machine learning applications. Many survey-based research articles (Koçer & Arslan, 2010; Lu et al., 2015; Niu, Liu, Wang, & Song, 2020; Pan & Yang, 2009; Weiss, Khoshgoftaar, & Wang, 2016; Zhuang et al., 2020) elaborate on its significance, development, and usefulness in multiple domains. Human activity recognition is one of those applications where transfer learning presence is essential in terms of cost and performance.

Shao, Zhu, and Li (2014) investigates the scope of transfer learning in the province of object recognition, image classification, and HAR. It emphasizes technical elaboration about cross-domain and cross-view feature representation transfer. Meanwhile, our paper focuses

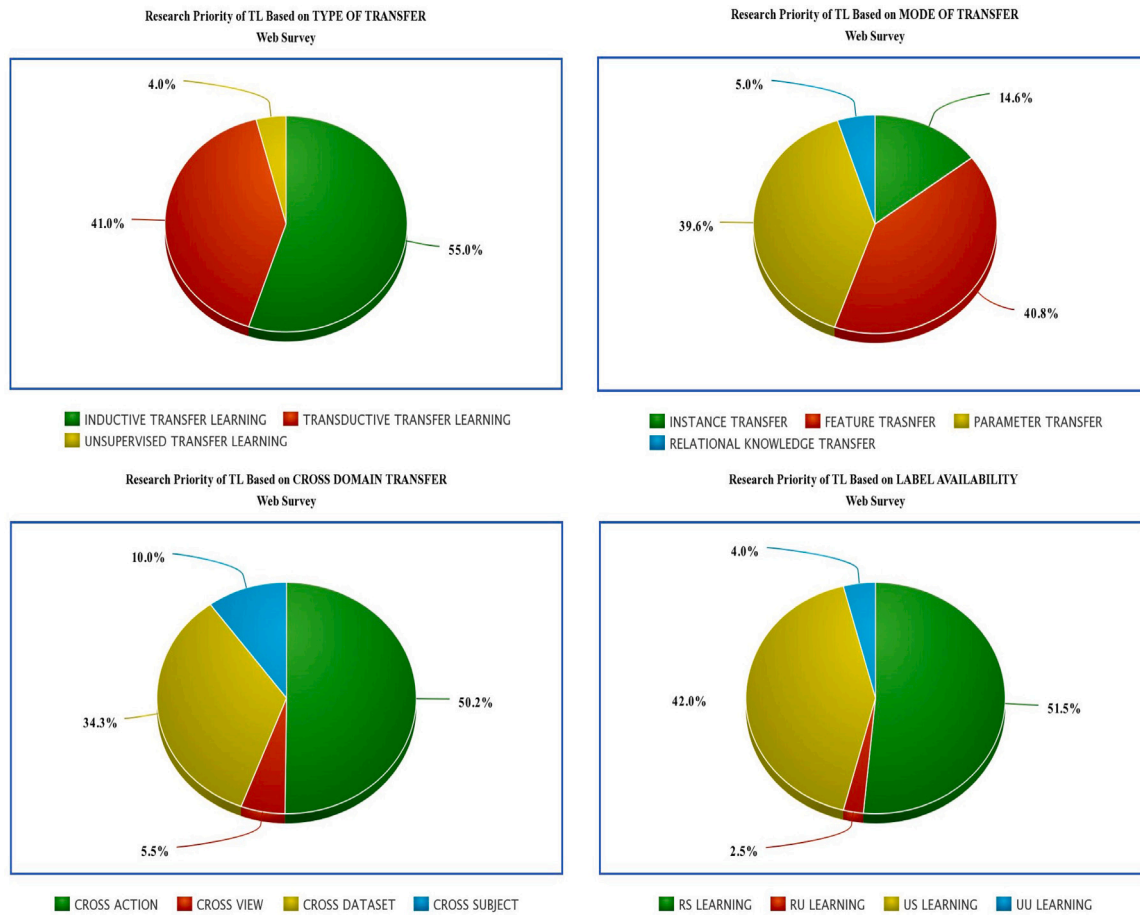


Fig. 1. Pi-chart representation for research priority related to transfer learning-based HAR. Here, RS, RU, US, and UU learning are used for the short-hand notation of Recognized-Supervised, Recognized-Unsupervised, Unrecognized-Supervised, and Unrecognized-Unsupervised learning respectively as described in Section 5.1.2. These above learning types are discussed in details in the Section 5.

only on HAR in the transfer learning platform. This paper does not spotlight the technical explanation. Instead, it provides an extensive accumulated database of related articles based on cross-domain and cross-view along with cross-action and cross dataset. Shao et al. (2014) convene information about knowledge-based model/classifier and its selection in visual categorization. However, our paper tries to assemble all significant knowledge regarding transfer learning-based HAR articles. Onofri, Soda, Pechenizkiy, and Iannello (2016) delivers a detailed study of different methods and techniques that help to integrate knowledge and reasoning into the HAR process. These promoted and contributed knowledge are based on statistical, probabilistic, and descriptive machine learning approaches. Instead, our paper contributes to understanding transfer learning-based HAR in state-of-the-art deep learning approaches.

Mabrouk and Zagrouba (2018) presents a detailed study of abnormal human behavior representation and modeling in intelligence surveillance technology. Feature extraction and description techniques, classification methods and frameworks, datasets, and performance metrics are the notable highlights of this paper. Our paper includes all these descriptions to a greater extent for recent research works but in the HAR domain using transfer learning techniques. Wang, Chen, et al. (2019) analyses and highlights various deep learning techniques encountered while approaching sensor-based activity recognition. It distinctively categorizes activity recognition articles based on different sensor modalities. Our paper tries to create a large and distinctive database of research articles considering various inferences, including sensor modality and deep learning techniques.

Nweke et al. (2018) provides an in-depth sketch of different deep learning methods and architecture associated with mobile and wearable sensor-based human activity recognition. Rather than representing detailed knowledge in the constrained mobile and wearable sensors domain, our paper tries to encapsulate human activity measured from all sensing platforms directly or indirectly linked with transfer learning. Wang, Cang, and Yu (2019) aims to furnish a comprehensive survey on wearable sensor modality-based HAR concerned with health care. However, our survey is limited to neither wearable sensors calibrated HAR nor healthcare-related HAR. Most sensor modalities and related sectors are considered for our globally pooled survey.

Dang et al. (2020) investigates sensor-based and vision-based human activity recognition by extensively classifying different HAR methodologies based on their pros and cons. Our paper carries similar content for action region recognition but in the transfer learning platform. Our survey does not abide by any constraints like sensor-based modeling, architecture-based modeling, classifier-based modeling, or dataset-based modeling, as seen in other surveys. This article combines all these models to forge a complete superficial package that will boost creativity for beginner and intermediate-level researchers.

Adama, Lotfi, and Ranson (2021) surveys different 2D and 3D vision-based HAR methods using transfer learning. Although the research domain is precisely matched with our paper, this survey more or less emphasizes 3D skeletal vision-based HAR. Dataset, feature extraction techniques, and ML/DL algorithms are discussed in Adama et al. (2021) focusing on skeletal action recognition. The key high-points and weaknesses of the HAR-related articles with compared to our paper are summarized in Tables 1 and 2.

Table 1

Review of existing surveys in HAR based on inductive transfer learning (2014–2021).

Author	Environment	Key highlights	Limitations
Shao et al. (2014)	Object /Image recognition, human activity recognition	Machine learning classifier-based knowledge transfer Depict cross-view/domain TL	No description about cross-action and cross-dataset TL No evaluation of HAR datasets Lack in information of HAR and learning diversity Analysis of challenges and future scopes are missing
Onofri et al. (2016)	HAR in video data	Knowledge-based machine learning approaches Public and private datasets used in knowledge-based HAR	No description about cross-data transfer Absence of statistical data comparison Learning and HAR diversity is not addressed Limited description about challenges and future scopes
Mabrouk and Zagrouba (2018)	Abnormal behavior in surveillance technology	Modeling frameworks and classification methods for abnormal behavior recognition Analyzes datasets and performance evaluation results	Absence of transfer learning techniques Challenges and future scopes are addressed Sensor diversity and pre-processing related information is missing
Wang, Chen, et al. (2019)	Sensor-based HAR	Deep learning techniques used in sensor-based HAR Segregate articles based on sensor modality Challenges ad future scopes are well addressed	Types of cross-data transfer is missing No categorization of HAR datasets Learning and HAR diversity is not addressed

Table 2

Review of existing surveys in HAR based on inductive transfer learning (2014–2021).

Author	Environment	Key highlights	Limitations
Nweke et al. (2018)	Mobile and wearable sensor-based HAR	Automatic feature extraction from mobile and wearable sensor data Different Deep learning framework for HAR Challenges and future scopes in Sensor-based HAR	Not based on transfer learning techniques Absence of cross data and statistical data analysis Learning and HAR diversity is not addressed
Wang, Cang, and Yu (2019)	HAR in healthcare	Focus on wearable sensor type, platform and placement Time/frequency/hybrid domain feature selection methods for Multi-modal features and their performances	Transfer learning technique is not included Only consider wearable data diversity Hardly any depiction about challenges and future directions
Dang et al. (2020)	Sensor and vision based HAR	Pre-processing and feature engineering methods Learning algorithm for sensor and vision based HAR Information about HAR datasets and its diversity	Transfer learning technique is not considered Lack in statistical data interpretation Limited related article consideration
Adama et al. (2021)	Vision-based HAR	3-D activity model for human skeleton data Evaluation for RGB+D HAR datasets	Cross-data transfer technique is not considered Limited examples for learning algorithms Only consider skeletal data
(Our paper)	Wearable, ambient, and Vision sensor-based HAR	Extend TL for skeleton data Our paper addresses all the above issues along with key highlights	Except in depth discussion of learning algorithms

3. HAR datasets

The HAR dataset evolves over time due to variations in content and technological advancements. By considering the quality and the number of videos in HAR datasets, we can segregate them into two groups (i) ancient age datasets and (ii) modern age datasets. Based on their popularity and use, a few have become state-of-the-art datasets. Modern datasets have more detailed specifications than old datasets. Some of the specifications can be physically distinguishable, whereas others cannot. Frame per second (fps), action count, actor count, video count per action, modality, resolution, annotation mode, and viewpoint are some physically accountable features. In contrast, light illumination and occlusion are physically indistinguishable features. The quality of

a dataset is determined by the factors listed above. A higher-quality dataset allows for the exploration of more composite models, whereas a complex dataset allows for examining the model's generalizability and resilience. We statistically analyze the average number of classes per dataset and average number of videos per dataset in consecutive years from 2009 to 2020 and are represented as shown in Fig. 2. Several factors help analyze a dataset and decide the group from where that dataset should belong. We can cluster HAR datasets into four levels of activities after reckoning the type of action, sensing modality, viewpoints, and the nature of the data. These are gesture level activity, atomic level activity, interaction level activity, i.e., Human-Human level (H-H) and Human-Object level (H-O), and group level activity which are summarized in Table 3. The ancient age datasets related

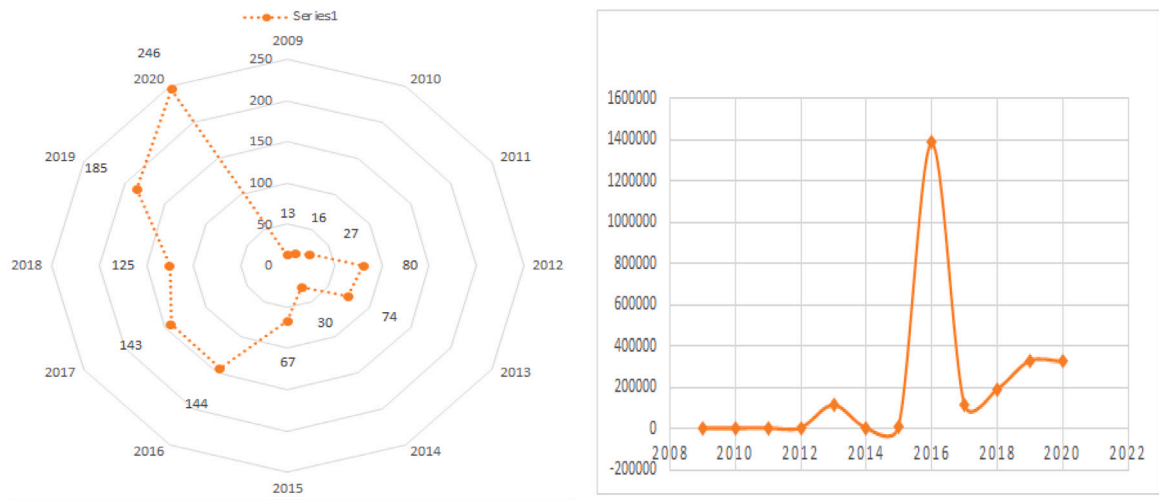


Fig. 2. Chronographic [2009–2020] representation of average number classes per dataset (left); Chronographic [2009–2020] representation of average number of videos per dataset (right).

Table 3		
Type of actions in human activity recognition.		
Type	Definition	Description
Gesture level activity	Purposive human body movement that convey some meaning or idea	Performed within a short period of time Considered the elemental activity among the five groups EX. Hand-waving, facial expressions, eye movements, and head shakes
Atomic level activity	Action comprising a sequence of interrelated gestures	Performed only by one actor without the involvement of any subject or object Sometimes referred to as a solo activity EX. Knocking, swimming, walking, jogging, and running
Interaction level activity	Results of interaction established between two agents	First agent must be a human, and the other one may be a human or an object If the second agent is an object, the action is called Human-Object interaction EX. cooking activities, kicking, hammering, eating, drinking, throwing, and sports activities like weight-lifting, batting and bowling If the second agent is a human, the action is called Human-human interaction EX. Hugging, wrestling, and shaking hands
Group level activity	Composite activity that require interaction between multiple persons and/or multiple objects	Involves many sequences of gestures, actions, and interactions EX. Group study, a cricket match, a group discussion, and a presentation

to these activity levels are summarized in Tables 4. The modern age datasets related to these activities level are summarized in Tables 5–7 with specifications. We also summarize the advantages and limitations of distinct state-of-the-art HAR datasets, as shown in Tables 8–10.

4. Human activity recognition linked to transfer learning domain

HAR is a prominent area of research while talking about indoor Activity of Daily Living (ADL), outdoor ADL, gesture recognition, sports activity recognition, micro-gesture recognition (Crockett, O’Shea, & Khan, 2020; Khan, Crockett, O’Shea, Hussain, & Khan, 2021; Khan, Hussain, Kuru, & Al-Askar, 2020), human–object interaction, (Akbari & Jafari, 2019; Al-Naser et al., 2018; Ben-Ari, Nacson, Azulai, Barzelay,

& Rotman, 2021; Chen, Liu, Fioranelli, Ritchie, & Chetty, 2019; Feuz & Cook, 2015; Fu, Damer, Kirchbuchner, & Kuijper, 2021; Gan, Yang, Zhu, Zhao, & Zhuang, 2016; Gao, Zhang, & Xu, 2019; Hasan & Roy-Chowdhury, 2015; Ijjina & Mohan, 2014; Jasani & Mazagonwalla, 2019; Ji, Liu, Pang, & Li, 2020; Ji, Liu, Pang, Ouyang, & Li, 2020; Ji, Yang, Xu, & Shen, 2018; Kalouris, Zacharaki, & Megalooikonomou, 2019; Kviatkovsky, Rivlin, & Shimshoni, 2014; Li, Wong, Zhao, & Kankanhalli, 2018; Liang & Thomaz, 2019; Liu, Liu, Gan, Tan, & Ma, 2018; Luo, Zou, Hoffman, & Fei-Fei, 2017; Mallya & Lazebnik, 2016; Mishra, Pandey, & Murthy, 2020; Mishra et al., 2018; Perrett, Masullo, Burghardt, Mirmehdi, & Damen, 2021; Pikramenos et al., 2020; Qi et al., 2020; Roitberg, Al-Halah, & Stiefelhagen, 2017; Rokni & Ghasemzadeh, 2018; Sabater et al., 2021; Sanabria, Zambonelli, &

Table 4
Old-aged (2004–2013) datasets with specifications.

Dataset[Year]	FPS	Resolution	Classes/Subjects	Videos	Mode application
KTH[2004]	25	160 × 120	6/25	600	Solo Activity
Weizmann[2005]	50	180 × 144	10/9	81	Solo Activity
IXMAS[2006]	23	640 × 480	11/13	1148	Solo Multi-view Activity
Kodak	30	640 × 480, 320 × 240	25/100	1358	Bank Interaction
CASIA Action[2007]	25	320 × 240	15/26	1446	H-H Interaction
UCF Sports[2008]	10	720 × 480	10/-	184	H-O & Group Sports Action
Hollywood[2008]	24	400 × 300, 300 × 200	8/-	233	H-H & Group Movies Action
i3DPost Multi-View[2009]	25	1920 × 1080	13/8	104	Solo action & Interaction
UCF11[2009]	29.97	320 × 240	11/-	3040	Interaction & Group Sports Action
Hollywood2[2009]	24	400 × 300, 300 × 200	22/-	3669	Group Movies Action
UT-Interaction[2009]	30	720 × 480	6/-	160	Interaction Activity
BEHAVE[2010]	25	640 × 480	10/5	163	Group & Interaction Activity
MSR Action[2010]	15	640 × 480	20/10	4020	Group Sports & Gesture Activity
Olympic Sports[2010]	–	–	16/-	800	Group Sports Action & Interaction
High-Five[2010]	–	–	5/-	300	H-H Interaction
RPI-ISL Activity Dataset[2010]	30	1920 × 1080	28/-	124	Solo & H-O Interaction (ADL)
CAD-60[2011]	30	640 × 480	12/4	60	Solo & H-O Interaction (ADL)
RGB-D HuDaAct[2011]	30	640 × 480	12/30	1189	Solo & H-O Interaction (ADL)
VIRAT(2.0)[2011]	30	1080 × 720	12/-	4 h	Solo & Group Surveillance
HMDB51[2011]	30	320 × 240	51/-	6766	H-H , H-O Interaction
UCF50[2011]	25	320 × 240	50/-	6676	Sports Group Action

Table 5
Modern age datasets with specifications.

Dataset[Year]Link	FPS/Resolution	Classes/Subjects	Videos	Activity level
UTKinect-Action[2012]/(Xia, Chen, & Aggarwal, 2012)	30/320 × 240	10/10	200	Atomic & H-O Level
Act4[2012]/(Cheng, Qin, Ye, Huang, & Tian, 2012)	30/640 × 480	14/24	6844	H-O level
ASLAN[2012]/(Kliper-Gross, Hassner, & Wolf, 2011)	-/-	432/-	3631	H-O/H-H/Group level
MP-II Cooking[2012]/(Rohrbach, Amin, Andriluka, & Schiele, 2012)	29.4/1624 × 1224	65/12	44	H-O level
UCF-101[2012]/(Soomro, Zamir, & Shah, 2012)	25/320 × 240	101/-	13,320	H-O/Group level
SBU Kinect Interaction[2012]/(Yun, Honorio, Chattopadhyay, Berg, & Samaras, 2012)	15/640 × 480	7/7	300	H-O level
G3D[2012]/(Bloom, Makris, & Argyriou, 2012)	25/640 × 480	20/10	210	Atomic/H-O level
BIT-Interaction[2012]/(Kong, Jia, & Fu, 2012)	30/320 × 240	8/-	400	H-H level
DML Smart Action[2013]/(Amiri, Pourazad, Nasiopoulos, & Leung, 2013)	30/2HD+1VGA	12/16	932	Atomic/H-O level
MSR Action 3D[2013]/(Li, Zhang, & Liu, 2010)	15/640 × 480	20/10	402	Atomic/H-O level
50-Salads[2013]/(Stein & McKenna, 2013)	30/640 × 480	10/27	966	H-O level
Berkeley MHAD[2013]/(Ofli, Chaudhry, Kurillo, Vidal, & Bajcsy, 2013)	22/640 × 480	11/12	660	Atomic/H-O level
CAD-120[2013]/(Sung, Ponce, Selman, & Saxena, 2012)	30/320 × 240	10/4	120	Atomic/H-O level
Hollywood 3D[2013]/(Hadfield & Bowden, 2013)	24/1920 × 1080	14/-	650	H-O/H-H level
JHMDB[2013]/(Jhuang, Gall, Zuffi, Schmid, & Black, 2013)	30/320 × 240	21/-	1.1 K	Atomic/H-O level
YouCook[2013]/(Das, Xu, Doell, & Corso, 2013)	-/-	10/-	88	H-O level
YouTube Sports 1M[2013]/(Karpathy et al., 2014)	-/227 × 227	487/-	11,33,158	H-O/Group level
Thumos' 14[2014]/Thumos14	-/-	101/-	18,000	Atomic/H-O/Group level
UWA3D Multi-View[2014]/(Rahmani, Mahmood, Du Huynh, & Mian, 2014)	30/640 × 480	30/10	900	Atomic/H-O level
Composable Activity[2014]/(Lillo, Soto, & Carlos Niebles, 2014)	30/640 × 480	16/14	693	Atomic/H-O level
Northwestern-UCLA[2014]/(Wang, Nie, Xia, Wu, & Zhu, 2014)	30/640 × 480	10/10	1475	Atomic/H-O level
Breakfast Action[2014]/(Kuehne, Arslan, & Serre, 2014)	15/320 × 240	10/52	2 K	H-O level
LIRIS[2014]/(Wolf et al., 2014)	25/640 × 480, 720 × 576	10/21	828	H-O/H-H level
IM Daily Depth[2014]/(Jalal, Kamal, & Kim, 2015)	30/640 × 480	15/15	45	Atomic/H-O level
Human Manipulation[2014]/(Pieropan, Salvi, Pauwels, & Kjellström, 2014)	30/640 × 480	6/8	140	H-O Level

Ye, 2021; Shen, Yeung, Hoffman, Mori, & Fei-Fei, 2018; Soleimani & Nazerfard, 2021; Wang & Chen, 2017b, 2020; Wang, Jiang, et al., 2018; Xu, Hospedales, & Gong, 2016, 2017; Xu et al., 2020; Zhang, Han, Tang, Hu, & Jiang, 2017; Zhang, Li, & Ogunbona, 2017; Zhu, Long, Guan, Newsam, & Shao, 2018) and human–human interaction (Antol, Zitnick, & Parikh, 2014; Chen, Wang, Huang, & Yu, 2019; Gan, Yang, et al., 2016; Gao et al., 2019; Hasan & Roy-Chowdhury, 2015; Ijjina & Mohan, 2014; Jasani & Mazagonwalla, 2019; Kodirov, Xiang, Fu, & Gong, 2015; Kviatkovsky et al., 2014; Liu et al., 2018; Luo et al., 2017; Mallia & Lazebnik, 2016; Mishra et al., 2018; Pauly, Agboh, Hogg, & Fuentes, 2021; Perrett et al., 2021; Roitberg et al., 2017; Tran, Bourdev, Fergus, Torresani, & Paluri, 2015; Wang & Chen, 2017b; Xu, Hospedales, & Gong, 2016, 2017; Yang, Mettes, & Snoek, 2021; Zhang, Han, et al., 2017; Zhu et al., 2018; Zou et al., 2018). Some of the prominent activity types relevant to this survey are tabulated in Tables 11 and

12. To understand the daily routine activity, we need a systematic tool that can differentiate and understand different activities. Under diverse situations, the activity recognition algorithm should perform more par than average to establish a well-performing recognition setup.

These days, transfer learning (Coskun et al., 2021) touches many fields of machine learning, basically, in computer vision tasks (Seyfioglu, Erol, Gurbuz, & Amin, 2018; Shrestha et al., 2019), where supervised learning is more prevalent in most of the research papers on bulky labeled datasets like ImageNet (Sani, Wiratunga, Massie, & Cooper, 2018). Five different modules fudge together to form the framework of human activity recognition, as shown in Fig. 3. These five modules are activity, sensing, transduction, approach, classification, and performances summarized distinctively below.

Table 6

Modern age datasets with specifications (Cont').

Dataset[Year]Link	FPS/Resolution	Classes/Subjects	Videos	Activity Level
MOBISERV-AIIA[2014]/(Iosifidis, Marami, Tefas, Pitas, & Lyroudia, 2015)	15/640 × 480	13/12	384	Atomic/H-O level
Watch-n-Patch[2015]/(Wu, Zhang, Savarese, & Saxena, 2015)	30/1920 × 1080, 512 × 424	21/7	458	Office & H-O level
G3Di[2015]/(Bloom, Argyriou, & Makris, 2014)	30/640 × 480	15/12	574	H-O/Group level
UTD_MHAD[2015]/(Chen, Jafari, & Kehtarnavaz, 2015)	30/640 × 480, 320 × 240	27/8	861	Atomic/H-O level
SYSU-3D HOI[2015]/(Hu, Zheng, Lai, & Zhang, 2015)	30/-	12/40	480	H-O level
ActivityNet[2015]/(Caba Heilbron, Escorcia, Ghanem, & Carlos Nieves, 2015)	30/1280 × 720	203/-	27,801	H-O level
THUMOS'15[2015]/(Gorban et al., 2015)	-/-	102/-	23,500	Atomic/H-O/Group level
MPII Cooking-2[2015]/(Rohrbach et al., 2016)	29.4/1624 × 1224	67/30	14,105	H-O level
SVW[2015]/(Safdarnejad et al., 2015)	1/1280 × 720	74/-	700 K	H-O/Group level
Crepe[2015]/(Lee, Ognibene, Chang, Kim, & Demiris, 2015)	-/-	9/3	53	H-O level
ChaLearn Action[2015]/(Escalera et al., 2014)	15/480 × 360	11/17	235	H-H level
FCVID[2015]/(Jiang, Wu, Wang, Xue, and Chang	-/-	239/-	91,223	H-O/Group level
OAD[2015]/(Li et al., 2016)	8/640 × 480	10/-	59	H-O level
M2I[2016]/(Liu, Xu, et al., 2016)	30/320 × 240	22/22	1760	H-H/H-O Interaction
NTU RGB+D 60[2016]/(Shahroudy, Liu, Ng, & Wang, 2016)	30/1920 × 1080, 512 × 424	60/40	56,880	Atomic/H-O/H-H level
MERL Shopping[2016]/(Singh, Marks, Jones, Tuzel, & Shao, 2016)	15/1920 × 1080	5/32	96	H-O level
Charades[2016]/(Sigurdsson et al., 2016)	-/-	157/-	9848	Atomic/H-O Interaction
YouTube 8M[2016]/(Abu-El-Hajja et al., 2016)	1/-	480/-	82,64,650	H-O/Group level
Vlog[2017]/(Fouhey, Kuo, Efron, & Malik, 2018)	154/660 × 1183	30/-	114 K	Atomic/H-O level
Kinetics400[2017]/(Kay et al., 2017)	-/658 × 1022	400/-	3,06,245	H-H/H-O Level
Kinetics600[2017]/(Carreira, Noland, Banki-Horvath, Hillier, & Zisserman, 2018)	-/658 × 1022	600/-	4,95,547	H-H/H-O level
A2D[2017]/(Xu, Xiong, & Corso, 2017)	-/-	14/-	3813	Atomic/H-O level
AVA[2017]/(Gu et al., 2018)	1/451 × 808	80/-	230 K	Atomic/H-O level
DALY[2017]/(Weinzaepfel, Martin, & Schmid, 2016)	1/1290 × 790	10/-	3.6 K	Atomic/H-O Level

Table 7

Modern age datasets with specifications (Cont').

Dataset/Year/Link	FPS/Resolution	Classes/Subjects	Videos	Activity Level
Multi-Thumos[2017]/(Yeung et al., 2018)	-/-	65/-	38 690	H-O/Group level
DAHLIA[2017]/(Vaquette, Orcesi, Lucat, & Achard, 2017)	15/1920 × 1080, 512 × 424	7/44	51	Atomic/H-O level
Okutama Action[2017]/(Barekatin et al., 2017)	30/3840 × 2160	12/9	43	Aerial H-H/H-O level
PKU-MMD[2017]/(Liu, Hu, Li, Song, & Liu, 2017)	30/1920 × 1080, 512 × 424	51/66	20000	H-O/H-H level
Something-SomethingV1[2017]/(Goyal et al., 2017)	12/84 × 84	174/1133	1,08,499	H-O level
Something-SomethingV2[2018]/(Goyal et al., 2017)	12/96 × 96	174/1133	2,20,847	H-O level
PRAXIS[2018]/(Negin, Rodriguez, Koperski, Kerboua, González, Bourgeois, Chapoulie, Robert, & Bremond, 2018)	-/960 × 540, 512 × 424	29/60	830 mins	Gesture/H-O level
Moments in Time[2018]/(Monfort et al., 2019)	25/340 × 256	339/-	8,02,264	H-O level
Stair Action[2018]/(Yoshikawa, Lin, & Takeuchi, 2018)	60/256 × 256	100/-	102 462	H-H/H-O level
MLB-YouTube[2018]/(Piergiovanni & Ryoo, 2018)	60/-	20/-	4290	H-O/Group level
SoccerNet[2018]/(Zhou, Xu, & Corso, 2018)	25/1280 × 720	3/-	6637	H-O/Group level
YouCook2[2018]/YouCook2	-/-	89/-	2000	H-O level
Multitask-AQA[2018]/(Parmar & Morris, 2019)	-/-	16/-	1412	H-O level
NTU RGB+D 120[2019]/(Liu, Shahroudy, et al., 2019)	30/1920 × 1080, 512 × 424	120/106	1,14,480	Atomic/H-O/H-H level
Diving48[2019]/(Kanojia, Kumawat, & Raman, 2019)	-/-	48/-	18 404	H-O level
HACS[2019]/(Zhao, Torralba, Torresani, & Yan, 2019)	-/-	200/-	1.5 M	Atomic/H-O/Group level
HAA-500[2019]/(Chung, Wu, Yang, Tai, & Tang, 2021)	-/1080 × 720	500/-	10 000	Atomic/H-O/Group level
MECCANO[2019]/(Ragusa, Livatino, Furnari, & Farinella	12/1080 × 720	61/20	20	H-O level
Kinetics-700[2020]/(Carreira, Noland, Hillier, & Zisserman, 2019)	-/658 × 1022	700/-	650 K	H-H/H-O level
MOD20[2021]/(Perera, Law, Ogunwa, & Chahl, 2020)	29.97/720 × 720	20/-	2324	H-O/Group level
EduNet[2021]/(Sharma, Gupta, Kumar, & Mishra, 2021)	30/1280 × 720	20/-	7851	H-O Level
TAD-08[2021]/(Gang et al., 2021)	-/720 × 576	8/-	2048	H-O level
Win-Fail[2021]/(Parmar & Morris, 2022)	-/1080 × 720	4/-	1634	Atomic/H-O level
IMU-based Gait[2022]/(Topham, Khan, Al-Jumeily, Waraich, & Hussain, 2022)	-/-	240/30	120,000	Atomic Level

4.1. Activity

People perform activities all the time, from off-bed to on-bed and even during sleep. Some human activities are performed individually, and some are realized with others' help. Based on the mode of performance, these activities can be button-downed into three classes: (i) Single-person activity (Ben-Ari et al., 2021; Li et al., 2018; Sun, Kong, Wang, Li, Yin, & Luo, 2021; Zhang et al., 2020), multi-person activity (Ji, Liu, Pang, & Li, 2020; Zhu et al., 2021), and group activity (Dwivedi, Gupta, Mitra, Ahmed, & Jain, 2019; Tran et al., 2015; Xu, Zhu, Wong, & Fang, 2016). Quantifying personal roles and linking these role descriptions to inference algorithms are the fundamental challenges we usually face during a group activity. We can categorize these activities as Indoor ADL, outdoor ADL, gesture, and cooking

activities under single-person or multi-person activity. Sports segments can be grouped under single-person or group activities. Tables 11 and 11 categorize various transfer learning-based activities along with their related paper links.

4.2. Sensing

Advancement in material science engineering helps the researcher develop large-scale semiconductor processing techniques to design new methods for transducing physical phenomena into easily processable electrical output. A sensor module helps recognize an event and its environmental changes and convey it to the control unit. Over the years, many advanced sensors have been introduced which can calibrate a wide range of physical, chemical, and biological properties. According

Table 8
Advantages and limitations of state-of-the-art datasets.

Datasets	Advantages	Limitations
Hollywood[2008]	• First dataset on movie scene classification • Individual variations in peoples' expression, posture, motion and clothing • Environmental variation in illuminations and scene dynamics • Provide dynamics perspective effects, occlusion, and camera motion • Automatic annotation using script-to-video alignment	• Actions are captured in controlled and simplified setting • Low resolution video frames • Only 8 classes of movie scenes
Hollywood2[2009]	• Provide more action classes (10) in addition with scene classes (12) • Provide variations and dynamics same as Hollywood • Comparative more realistic and challenging setting compare to Hollywood	• Low resolution video frames • Still unable to provide actions close to natural settings
OPPERTUNITY[2010]	• Naturalistic human activities in a multi-modal (10)environment • Well-suited for investigating sensor fusion methods and ontology-based reasoning • Dataset of complex, interleaved and hierarchical naturalistic activities • Context recognition in opportunistic sensor configurations • Deployed multiple wireless and wired networked sensor systems from different origins	• Challenging sensor setup with respect to data acquisition, synchronization, and curation
Olympic Sports[2010]	• First benchmark Olympic sports dataset • Annotated by crowd-sourcing platform i.e., Amazon Mechanical Turk (AMT)	• Small dataset having only 16 classes • No ground truth is provided except simple action labels

Table 9
Advantages and limitations of state-of-the-art datasets.

Datasets	Advantages	Limitations
HMDB-51[2011]	• Large video dataset of 51 actions with 6849 video clips • Annotated based on space-time interest point (STIP) • Also introduces diversity in terms of occlusion, camera motion, camera view-point, actors, number of people involved in the action	• Camera motion interfere the local motion computation • Low resolution dataset
UCF-101[2011]	• Large scale action dataset of 101 action • Challenging due to being collected in the wild	• Low quality video clips • Ground-truth provided in form of bounding boxes and action annotations
YouTube Sports-1M[2013]	• Benchmark event recognition dataset • Very large sport activity dataset having 487 classes • Multi-class labeling • Dataset id fine-grained by the leaf level • Automatic annotated by analyzing the text metadata surrounding the videos	• Weakly annotated dataset • Variable setting dataset as taken from YouTube videos
ActivityNet[2015]	• Large activity dataset with high resolution videos • Variety in the video data represents a real world challenge for existing algorithms • Compare to other datasets at this age, it contains variety in terms of activity diversity and richness of taxonomy • Comprise rarely considering activity	• Less useful now-a-days due to background variations • Class imbalance problem due to rare occurring classes

to the activity recognition point of view, we differentiate these sensors into two classes:

1. In HAR, On-body or wearable sensors is installed on the human body and record various activity-related signals. [Tables 13](#) enlists the frequently used on-body sensors and respective related links.
2. Sensors that are not installed on the human body but installed in the performing environment are called Off-body or ambient sensors. [Tables 14](#) comprises off-body sensors and respective related articles.

4.3. Transduction

Sensor acts as the nervous system whereas, the algorithms and processing units serve as the brain of activity recognition technology. However, the transduction unit interlinks the sensor output and the processing unit input after suitable signal conversion. Transduction is the process of converting the original sensed signal to an appropriate and easily processable form. The state of outgoing signals is highly dependent on the adjoining processing unit. The sensing unit and the transduction unit fuse together most of the time, called a transducer. Nevertheless, sometimes it is convenient for the manufacturer

Table 10
Advantages and limitations of state-of-the-art datasets.

Datasets	Advantages	Limitations
Something-Something V2[2017]	• Large scale dataset having 220.8K video clips distributed over 172 classes • Avg. 620 videos per classes	• Learning the action only through vision but not by interaction to the world • Presence of dataset bias using different cues • Still follows manual labeling strategy
NTU RGB+D 120[2019]	• Labeled by crowdsourcing services like AMT • Dataset comprises labeled video data with natural language descriptions • Low entropy of the data distribution can be exploited • Largest 3D dataset with depth features • Collect four major data modalities i.e., the depth maps, the 3D joint information, the RGB frames, and the infrared sequences • Real-time skeleton-based early action recognition • Can be used for one-shot 3D activity analysis and mutual activity analysis • Provide environment for cross-modal transfer learning • High resolution RGB videos (1920×1080) • Cross-subject, cross-view, and cross-environment activity analysis • Heterogeneous feature fusion analysis • Large scale high quality dataset	• Generated in constrained lab environment • Not suitable for natural action recognition tasks • Still follows manual labeling strategy
Kinetics[2017–2020]	• Diverse range of human action • Geo-diversified dataset Due to sound signal incorporation, suitable for multi-modal analysis • Presence of camera motion, illumination variations, shadows and background clutter • Presence of performance variation like pattern, clothing, age, gender and camera clothing	• Not suitable for finding action classes that progress over time • Manual labeling strategy • Little evidence of gender biasing in certain application

to place the transduction device outside the sensor termed out-built transduction.

4.4. Approach

The intent to improvise numerous deciding factors like compatibility, portability, cost, efficiency, and accuracy has fueled researchers in the path of innovation. The high-tech world appears with a new face every day by adding advanced technical features that accomplish a more efficient and robust device than its previous version. Multiple approaches and algorithms are developed for more efficient and accurate activity recognition in the course of the journey. Some of them are obsolete, some have restricted use, and some of them have become state-of-the-art methods. The probabilistic and statistical modeling approach is introduced in activity recognition to explicitly represent uncertainty in reasoning about the performer's actions, plans, and goals. History says that many probabilistic and statistical models are adopted in solving activity recognition problems, such as Decision Trees (DT), Naïve Bayes Classifier (NBC), and HMM.

The probabilistic model has failed to bring so in a real environment because of its dependency on other independent inferences. Machine learning is the most successful activity recognition framework, where popular deep learning is a subset of this learning technique. K-Nearest Neighbors (KNN) (Xu, Zhu, et al., 2016), Bayes classifier (Vondrick et al., 2016) Minimal Learning Machine (MLM) (Karpathy et al., 2014), Support Vector Machine (SVM) (Bhattacharya et al., 2014; Oh et al., 2021), Extreme Learning Machine (ELM) (Wang, Chang, et al., 2018), and Multi-layer Perceptron (MLP) (Liu & Ferrari, 2017) are some popular ML techniques practiced in inactivity recognition platforms.

However, nowadays, deep learning techniques' popularity is sky-high as they can bring out the highest-ever success rate in real-world

environments. The deep learning technique sets a platform for automatic feature selection and learning compared to machine learning handcrafted feature extraction. Some of the frequently used deep learning techniques are Deep Neural Network (DNN) (Dwivedi et al., 2019; Feng & Duarte, 2019), Convolutional Neural Network (CNN) (Rodriguez et al., 2017; Yang et al., 2012), autoencoder (Jiang et al., 2015), Restricted Boltzmann Machine (RBM) (Diethe, Twomey, & Flach, 2015; Wan et al., 2015), Recurrent Neural Network (RNN) (Dwivedi et al., 2019), (Bagaveyev & Cook, 2014), and generative Adversarial Network (GAN). Tables 16 and 26 enlist the significant articles based on the aforementioned deep learning techniques.

4.5. Performance measures

In a predictive task, every user should understand the acceptance level of predictive results. To get this, we need to find out the correct prediction probability. It can be interpreted as the ratio between all the accurate predictions and the total prediction count. Intuitively, users of a prediction model would like to understand how likely they may rely on the prediction outcomes. However, the researcher developed a set of statistical performance parameters for the quantitative description of predictive performances in various aspects under multiple circumstances. While talking about class, we come across two kinds of labels; truth labels and prediction labels. Ground truth labels are the actual class labels the samples belong to, and prediction labels are model-predicted tags after evaluation, either during validation or testing. According to these definitions, if both ground truth and predictive labels are positive, they are referred to as a true positive sample, and negative is referred to as a true negative sample. False-positive if the ground truth label is negative, but the prediction label is positive, and false negative if the ground truth label is positive, but the prediction

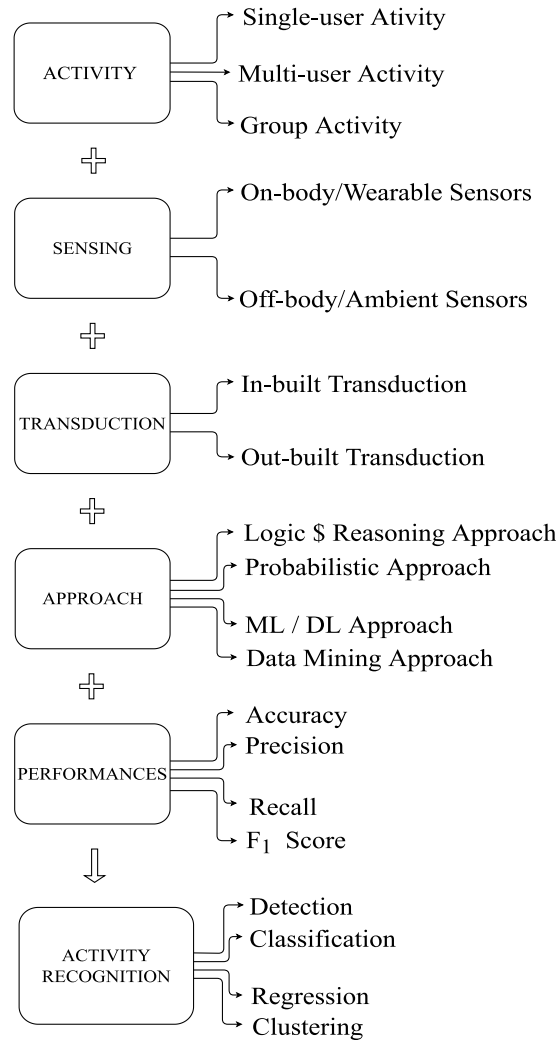


Fig. 3. Diagrammatic representation of various steps involved in HAR.

label is negative. While talking about activity recognition, the most commonly used performance measures are given below.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \times 100 \% \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \times 100 \% \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \times 100 \% \quad (3)$$

$$F_1 \text{ score} = \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Apart from these measures, two other measures are also mentioned in different articles, i.e., the Mean Levenshtein Distance (MLD) score and mean average precision. Levenshtein Distance (LD) helped to evaluate the performance between ground truth labels and predicted labels frequently used in gesture recognition [89]. The MAP score is the mean AP calculated over all classes and overall IoU thresholds depending on the challenges. Table 15 helps to find the related articles based on these performance measures.

5. Variants of transfer learning in human activity recognition

In activity recognition, the transfer learning type can be represented in terms of three-mode framework based on the data and environment settings as shown in Fig. 4. These are (i) data representation centric TL, (ii) data analysis centric TL, and (iii) data and environment centric TL.

5.1. Data representation centric TL

This mode of transfer learning depends either on the label information or on the space information of the source and target data. Based on data representation, it is further divided into three sub-groups. (i) TL based on machine learning settings, (ii) TL based on label information settings, and (iii) TL based on space information settings.

5.1.1. TL based on machine learning settings

Based on machine learning settings, transfer learning can be used in inductive, transductive, or unsupervised platforms. These are described below:

1. **Inductive transfer learning:** Some target domain label data is required to induce the conditional probability distribution function irrespective of label data availability in the source domain; Tables 16–20 discuss many research articles based on inductive transfer learning. Self-taught TL (Mumtaz, Sargano, & Habib, 2018) and multi-task learning have been placed under the radar of inductive transfer learning according to the presence of source domain label data.

Self-taught Learning It deals with the unlabeled source domain data to improve the performance of supervised learning tasks more efficiently. It manipulates a more extensive unlabeled source database to find a common representation that helps to enhance the performance of labeled target domain data.

Table 11
Paper links of different classified activities related to transfer learning based HAR.

Activity	Related paper links
Outdoor ADL	Akbari and Jafari (2019), Al-Naser et al. (2018), Alexiou, Xiang, and Gong (2016), Antol et al. (2014), Ben-Ari et al. (2021), Brattoli, Tighe, Zhdanov, Perona, and Chalupka (2020), Cao, Ji, Cao, Chang, and Niebles (2020), Careaga, Hutchinson, Hodas, and Phillips (2019), Chang, Mathur, Isopoussu, Song, and Kawsar (2020), Chavarriaga, Bayati, and Millán (2013), Chen, Liu, et al. (2019), Chen, Wang, et al. (2019), de Souza, Gaidon, Cabon, and López (2017), Deng, Hua, Wang, Wang, and Zhou (2020), Deng, Zheng, and Wang (2014), Ding et al. (2018), Du, Farrahi, and Niranjan (2019), Du, He, and Jin (2018), Du, Jin, Song, and Dai (2020), Du, Jin, Song, Dai, and Li (2019), Fallahzadeh and Ghasemzadeh (2017), Feng and Duarte (2019), Fetter and Gross (2014), Fu et al. (2021, 2019), Gan, Lin, Yang, De Melo, and Hauptmann (2016), Gan, Lin, Yang, Zhuang, and Hauptmann (2015), Gan, Yang, et al. (2016), Gao et al. (2019), Guo et al. (2018), Haddad, Ghassab, Najjar, and Bouguila (2021), He, Li, and Jing (2019), Hong, Ramos, and Dey (2015), Hu, Sun, Saenko, and Sclaroff (2019), Iijina and Mohan (2014), Jain, Van Gemert, Mensink, and Snoek (2015), Jänicke, Tomforde, and Sick (2016), Jasani and Mazagonwalla (2019), Ji, Liu, Pang, and Li (2020), Ji, Liu, Pang, Ouyang, and Li (2020), Ji et al. (2018), Jiang et al. (2018), Kalouris et al. (2019), Karpathy et al. (2014), Kay et al. (2017), Ke, Bennamoun, An, Sohel, and Boussaid (2017), Khan and Roy (2018), Khan, Roy, and Misra (2018), Kodirov et al. (2015), Kviatkovsky et al. (2014), Lang et al. (2018, 2019), Laput, Ahuja, Goel, and Harrison (2018), Li, Jing, and He (2020), Li et al. (2018), Liu et al. (2018), Liu, Liu, Ma, Huang, and Dong (2019), Liu et al. (2014), Liu, Su, Nie, and Kankanhalli (2016), Liu, Xu, et al. (2015), Luo et al. (2017), Mallya and Lazebnik (2016), Mandal et al. (2019), Mazankiewicz, Böhm, and Bergés (2020), Memmesheimer, Theisen, and Paulus (2020), Mishra et al. (2020, 2018), Morales and Roggen (2016), Oh, Ashiquzzaman, Lee, Kim, and Kim (2021), Papagiannaki et al. (2019), Park, Javier, Moon, and Kim (2016), Pauly et al. (2021), Perrett et al. (2021), Pikramenos et al. (2020), Prabono, Yahya, and Lee (2021), Qi et al. (2020), Qin, Chen, Wang, and Yu (2019), Rahmani and Mian (2015), Rodriguez, Orrite, Medrano, and Makris (2017), Roitberg et al. (2017), Roitberg, Martinez, Haurilet, and Stiefelhaven (2018), Rokni and Ghasemzadeh (2018), Rokni, Nourollahi, and Ghasemzadeh (2018), Sabater et al. (2021), Saeed, Ozelebi, and Lukkien (2019), Saeedi, Ghasemzadeh, and Gebremedhin (2016), Saeedi, Sasani, Norgaard, and Gebremedhin (2018), Sanabria et al. (2021), Seyfioglu et al. (2018), Shen et al. (2018), Shi, Zhang, Xu, Cheng, and Pearce (2020), Sun et al. (2021), Sztylek, Stuckenschmidt, and Petrich (2017), Thaker and Krishnakumar (2017), Tran et al. (2015), Vondrick, Pirsivash, and Torralba (2016), Wan, Guo, and Li (2015), Wang, Chang, Shi, Yang, and Shen (2018), Wang and Chen (2017a, 2017b), Wang, Chen, Hu, Peng, and Philip (2018), Wang, Jiang, et al. (2018), Wang, Zheng, Chen, and Huang (2018), Xu, Hospedales, and Gong (2015, 2016, 2017), Xu et al. (2020), Xu, Ye, Zheng, Wang, Luwang, and Jiang (2018), Yang et al. (2021), Yang, Saleemi, and Shah (2012), Yao et al. (2016), Zhang, Han, et al. (2017), Zhang, Li, and Ogunbona (2017), Zhang, Li, and Rehg (2017), Zhang, Wang, Xiao, Zhou, and Liu (2015), Zhang et al. (2013, 2020), Zhang, Zhou, and He (2021), Zhou and Mu (2020), Zhu et al. (2018), Zhu and Newsam (2017), Zhu and Shao (2014a), Zhu et al. (2021), Zhu and Yang (2018), Zou et al. (2020, 2018)
Indoor ADL	Akbari and Jafari (2019), Al-Naser et al. (2018), Alam and Roy (2017), Alexiou et al. (2016), Bhattacharya, Nurmi, Hammerla, and Plötz (2014), Brattoli et al. (2020), Cao et al. (2020), Careaga et al. (2019), Chang et al. (2020), Chavarriaga et al. (2013), Chen, Wang, et al. (2019), Diethe, Twomey, and Flach (2016), Du, Farrahi, and Niranjan (2019), Feuz and Cook (2015, 2017), Gan, Lin, et al. (2016), Gan et al. (2015), Gao et al. (2019), Jain et al. (2015), Khan and Roy (2018), Kodirov et al. (2015), Liang and Thomaz (2019), Liu et al. (2018), Liu, Liu, et al. (2019), Mandal et al. (2019), Mazankiewicz et al. (2020), Mishra et al. (2020, 2018), Morales and Roggen (2016), Narui, Shu, Gonzalez-Navarro, and Ermon (2020), Papagiannaki et al. (2019), Perrett et al. (2021), Pikramenos et al. (2020), Prabono, Yahya, and Lee (2020), Prabono et al. (2021), Rey and Lukowicz (2017), Roitberg et al. (2017), Rokni and Ghasemzadeh (2017), Sanabria and Ye (2020), Shrestha et al. (2019), Shu, Bui, Narui, and Ermon (2018), Soleimani and Nazerfard (2021), Sun et al. (2021), Wan et al. (2015), Wang and Chen (2017a, 2017b), Wang, Zheng, et al. (2018), Wen, Indulska, and Zhong (2016), Wen and Zhong (2015), Xu et al. (2015), Xu, Hospedales, and Gong (2016, 2017), Ye (2018), Ying, Lin, Tseng, and Hsieh (2015), Zhang, Han, et al. (2017), Zhang et al. (2021), Zhu et al. (2018)

Table 12
Paper links of different classified activities related to transfer learning based HAR.

Activity	Related paper links
Gesture	Alam and Roy (2017), Cabrera, Sanchez-Tamayo, Voyles, and Wachs (2017), Cabrera and Wachs (2017), Chang et al. (2020), Chen, Wang, et al. (2019), Deng et al. (2020), Jiang, Zhang, Wu, Gao, and Zhao (2015), Karn and Jiang (2016), Kviatkovsky et al. (2014), Li et al. (2021), Lin, Ruan, Yu, and Wei (2015), Ma, Zhang, Wang, Qi, and Chen (2020), Morales and Roggen (2016), Rey and Lukowicz (2017), Rodriguez et al. (2017), Soleimani and Nazerfard (2021), Yamada, Sigal, and Raptis (2014), Yang et al. (2012), Zhang, Zhang, et al. (2017)
Sports	Alexiou et al. (2016), Ben-Ari et al. (2021), Brattoli et al. (2020), Cao et al. (2020), Careaga et al. (2019), Chang et al. (2020), Chen, Wang, et al. (2019), de Souza et al. (2017), Ding et al. (2018), Dwivedi et al. (2019), Fallahzadeh and Ghasemzadeh (2017), Fu et al. (2021), Gan, Lin, et al. (2016), Gan et al. (2015), Gan, Yang, et al. (2016), Gao et al. (2019), Hu et al. (2019), Jain et al. (2015), Jasani and Mazagonwalla (2019), Kalouris et al. (2019), Karpathy et al. (2014), Kay et al. (2017), Khan and Roy (2018), Kodirov et al. (2015), Kviatkovsky et al. (2014), Liu et al. (2018), Liu, Liu, et al. (2019), Liu, Mei, Zhang, Che, and Luo (2015), Liu et al. (2014), Liu, Su, et al. (2016), Liu, Wu, and Jia (2015), Luo et al. (2017), Mallya and Lazebnik (2016), Mandal et al. (2019), Mettes and Snoek (2017), Mishra et al. (2020, 2018), Pauly et al. (2021), Perrett et al. (2021), Prabono et al. (2021), Roitberg et al. (2017), Rokni et al. (2018), Saeedi et al. (2018), Sun et al. (2021), Sun, Shetty, Sukthankar, and Nevatia (2015), Tran et al. (2015), Vondrick et al. (2016), Wang, Chang, et al. (2018), Wang and Chen (2017a, 2017b), Xu et al. (2015), Xu, Hospedales, and Gong (2016, 2017), Xu et al. (2020, 2018), Xu, Zhu, et al. (2016), Zhang, Han, et al. (2017), Zhang et al. (2020, 2021), Zhu et al. (2018), Zhu and Newsam (2017), Zhu and Yang (2018), Zou et al. (2020, 2018)
Cooking	Coskun et al. (2021), Soleimani and Nazerfard (2021), Wan et al. (2015), Wang and Chen (2020), Zhu and Newsam (2017)

Table 13
Sensors used in transfer learning-based human activity recognition articles.

On-body sensor	Related paper links
Smart phone	Akbari and Jafari (2019), Deng et al. (2014), Diethe et al. (2016), Hong et al. (2015), Kalouris et al. (2019), Khan and Roy (2018), Khan et al. (2018), Liang and Thomaz (2019), Mazankiewicz et al. (2020), Qin et al. (2019), Saeed et al. (2019), Saeedi et al. (2016), Szttyler et al. (2017), Yao et al. (2016)
Smart watch	Akbari and Jafari (2019), Khan et al. (2018), Mazankiewicz et al. (2020), Saeed et al. (2019), Szttyler et al. (2017)
Accelerometer	Alam and Roy (2017), Bhattacharya et al. (2014), Chang et al. (2020), Chavarriaga et al. (2013), Chen, Wang, et al. (2019), Deng et al. (2020, 2014), Diethe et al. (2016), Ding et al. (2018), Du, Farrahi, and Niranjana (2019), Fallahzadeh and Ghasemzadeh (2017), Feng and Duarte (2019), Feuz and Cook (2017), Ijjina and Mohan (2014), Jänicke et al. (2016), Kalouris et al. (2019), Khan and Roy (2018), Khan et al. (2018), Liu, Wu, and Jia (2015), Mazankiewicz et al. (2020), Morales and Roggen (2016), Narui et al. (2020), Oh et al. (2021), Papagiannaki et al. (2019), Prabono et al. (2020, 2021), Qin et al. (2019), Rey and Lukowicz (2017), Rokni and Ghasemzadeh (2017, 2018), Rokni et al. (2018), Saeed et al. (2019), Saeedi et al. (2016, 2018), Sanabria et al. (2021), Soleimani and Nazerfard (2021), Szttyler et al. (2017), Wang, Chen, et al. (2018), Wang, Zheng, et al. (2018), Wen et al. (2016), Wen and Zhong (2015), Xu, Zhu, et al. (2016), Ying et al. (2015)
Gyroscope	Bhattacharya et al. (2014), Chang et al. (2020), Chen, Wang, et al. (2019), Deng et al. (2020, 2014), Diethe et al. (2016), Ding et al. (2018), Du, Farrahi, and Niranjana (2019), Fallahzadeh and Ghasemzadeh (2017), Feng and Duarte (2019), Feuz and Cook (2017), Jänicke et al. (2016), Kalouris et al. (2019), Khan and Roy (2018), Mazankiewicz et al. (2020), Morales and Roggen (2016), Oh et al. (2021), Papagiannaki et al. (2019), Prabono et al. (2020, 2021), Qin et al. (2019), Rey and Lukowicz (2017), Rokni and Ghasemzadeh (2017, 2018), Saeed et al. (2019), Saeedi et al. (2018), Soleimani and Nazerfard (2021), Szttyler et al. (2017), Wang, Chen, et al. (2018), Wang, Zheng, et al. (2018), Wen et al. (2016), Wen and Zhong (2015)
Magnetometer	Bhattacharya et al. (2014), Chang et al. (2020), Chen, Wang, et al. (2019), Ding et al. (2018), Du, Farrahi, and Niranjana (2019), Fallahzadeh and Ghasemzadeh (2017), Feng and Duarte (2019), Feuz and Cook (2017), Kalouris et al. (2019), Khan and Roy (2018), Morales and Roggen (2016), Papagiannaki et al. (2019), Prabono et al. (2020, 2021), Qin et al. (2019), Rey and Lukowicz (2017), Rokni and Ghasemzadeh (2017, 2018), Saeedi et al. (2018), Soleimani and Nazerfard (2021), Szttyler et al. (2017), Wang, Chen, et al. (2018), Wang, Zheng, et al. (2018), Wen et al. (2016)
Inertia	Bhattacharya et al. (2014), Chang et al. (2020), Chen, Wang, et al. (2019), Du, Farrahi, and Niranjana (2019), Feng and Duarte (2019), Feuz and Cook (2017), Kalouris et al. (2019), Khan and Roy (2018), Morales and Roggen (2016), Prabono et al. (2020, 2021), Rey and Lukowicz (2017), Rokni and Ghasemzadeh (2017), Rokni et al. (2018), Sanabria et al. (2021), Soleimani and Nazerfard (2021), Wang, Zheng, et al. (2018), Wen et al. (2016), Xu, Zhu, et al. (2016)
Microphone	Ijjina and Mohan (2014), Laput et al. (2018), Liang and Thomaz (2019), Szttyler et al. (2017), Ying et al. (2015)
Heart rate	Chen, Wang, et al. (2019), Kalouris et al. (2019), Sanabria et al. (2021)
MOCAP	Chen, Liu, et al. (2019), Du et al. (2020), Ijjina and Mohan (2014), Yamada et al. (2014)

The combined representation will utilize to recognize unlabeled activity in the target domain.

Multi-task Learning (MTL) MTL (de Souza et al., 2017; He et al., 2019; Ke et al., 2017; Lang et al., 2019; Lin et al., 2015; Liu, Mei, et al., 2015; Liu et al., 2014; Liu, Su, et al., 2016; Shrestha et al., 2019; Zhu & Newsam, 2017) is a class of transfer learning where a common shared platform concurrently learns multiple tasks. MTL can also be treated as inductive transfer learning where both source and target domain activity labels are available. Overfitting reduction due to shared representations, fast learning by leveraging auxiliary knowledge, and enhanced data efficiency are some advantages of MTL. MTL methods can be radared under two categories in deep learning. Hard parameter sharing and soft parameter sharing.

2. **Transductive transfer learning:** The target domain label information is missing in transductive TL. The source label data information is transferred from the source to the target domain to induce the conditional probability distribution function. **Tables 21–25** chronologically summarize transductive TL-related research articles. Two different classes of transductive TL are briefed below.

Domain adaptation (DA) The model training of DA is based on source domain data distribution that helps to explore and learn a different but related target domain data distribution. The more intense the bonding between source and target data, the more successful the adaptation will be. Depending on the depth of adaptation, we can differentiate DA into two categories: (i) Shallow domain adaptation and (ii) Deep domain adaptation. Based on the type of transfer, we can also split DA into four categories: (i) Feature-based DA (Akbari & Jafari, 2019; Chang et al., 2020; Chen, Liu, et al., 2019; Chen, Wang, et al., 2019;

Ding et al., 2018; Du et al., 2020; Jänicke et al., 2016; Jiang et al., 2018; Lang et al., 2018; Li et al., 2020; Liang & Thomaz, 2019; Mazankiewicz et al., 2020; Mishra et al., 2018; Narui et al., 2020; Pfister et al., 2014; Prabono et al., 2020, 2021; Saeedi et al., 2016; Sanabria & Ye, 2020; Shu et al., 2018; Zhang, Li, & Ogunbona, 2017; Zhu & Shao, 2014a) (ii) Parameter-based DA (Kalouris et al., 2019; Liang & Thomaz, 2019; Prabono et al., 2020) (iii) Instance-based DA (Khan et al., 2018; Sanabria et al., 2021) (iv) Relational knowledge-based DA (Qin et al., 2019; Rey & Lukowicz, 2017).

Co-variate shift The underlying distribution of training data samples is generally the same as the testing data samples when both are taken from similar sensors and/or similar environmental settings. However, in a real-world scenario, this assumption does not hold good which causes performance degradation. Transfer learning helps to shift the variance distribution curve of one towards another to minimize the curve distance. By maximizing the co-variance distribution of training and testing input variables, we can enhance the model performance to a large extent.

3. **Unsupervised transfer learning:** No label information is present in either domain. Transfer of information takes place in an unsupervised manner. **Table 26** reviews some significant research articles based on unsupervised transfer learning.

5.1.2. TL based on label information setting

We can classify transfer learning based on the availability of the labeled data information in the source and target domain. For that, we refer to the terms supervised and unsupervised by scrutinizing the source domain label information. Meanwhile, we use recognized and unrecognized terms after detecting the target domain label information.

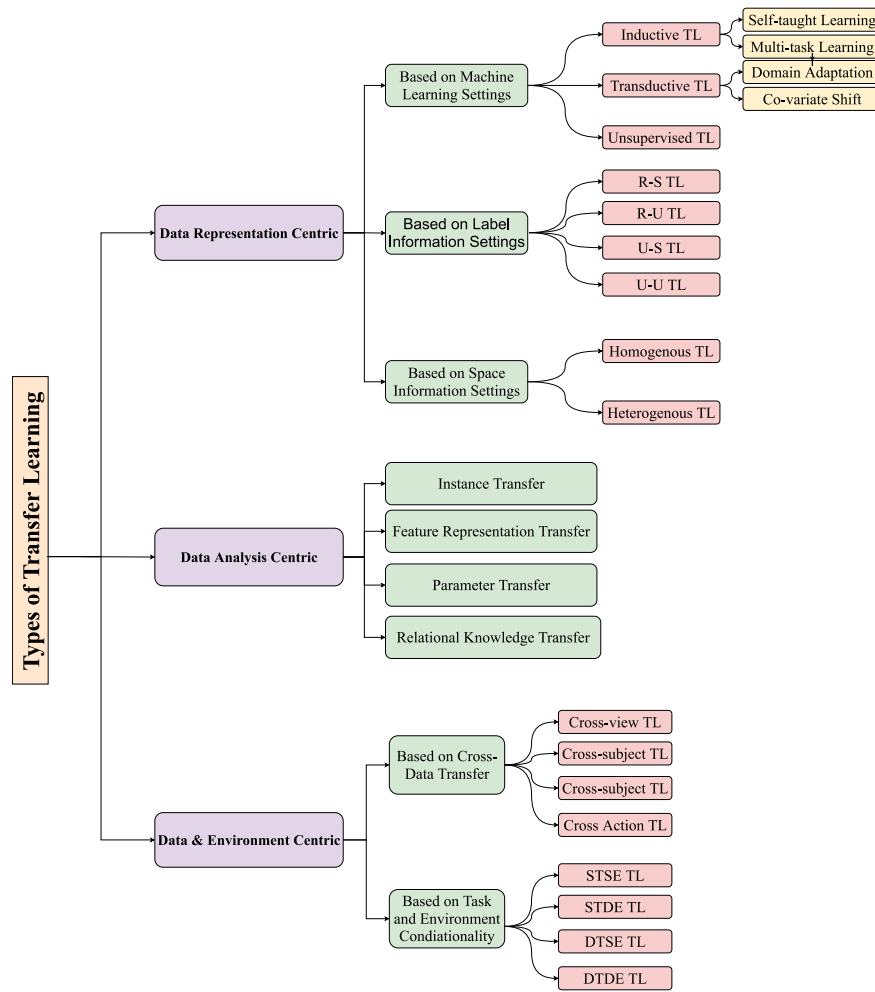


Fig. 4. Three modular representations of transfer learning related to human activity recognition.

By integrating the above two conditions, we propose four transfer learning techniques: (i) Recognized Supervised (RS) TL, (ii) Recognized Unsupervised (RU) TL, (iii) Unrecognized supervised TL (US), and (iv) Unrecognized unsupervised (UU) TL. After taking this hypothesis, RS transfer learning has evolved from both source and target domain labels available for model training. Likewise, we can refer to the transfer learning type as RU, US, and UU when labels are available only in the target domain, source domain, and neither domain, respectively. The above four techniques are adopted by many of the existing works, which are tabulated in Table 27. US transfer activity learning can also sometimes refer to transductive learning, where we are more interested in learning the relationship between the source and target domain instances. The Teacher-Learner model (Papagiannaki et al., 2019) is a more familiar framework while working with a UU activity transfer technique. Here, a teacher classifier is trained on the source domain data, followed by a learner classifier that is trained synchronously with the teacher to exploit target data. The learner's accuracy, which is nothing but the overall accuracy, depends on how excellent features a teacher can exploit during training. Therefore, much attention is given while modeling teacher architecture.

5.1.3. TL based on space information setting

Based on the availability of space information in the source and target data, TL is segmented into two sub-groups:

1. *Homogeneous transfer learning*: A learning technique can be called homogeneous when analogous features transfer from the source domain to the target domain. It can support all four data analysis transfer learning methods which are discussed in the below section. In activity recognition, the homogeneous transfer may be carried out through four methods, i.e., instance transfer, feature representation transfer, parameter transfer, and relational knowledge transfer method, as discussed in Section 5.2.
2. *Heterogeneous transfer learning*: Here, the features are completely different from source to target. It can only support the feature representation transfer method.

5.2. Data analysis centric TL

We may encounter three fundamental issues while analyzing TL-related data. "what to transfer" is one of them alongside "how to transfer" and "when to transfer." Based on which part of the knowledge is going to transfer, this mode of transfer learning can be segmented into four sub-groups:

5.2.1. Instance transfer

Instance transfer reuses the source data to train the target data by re-weighting the source instances governed by a pre-defined metric. It is frequently applied after verifying that the source feature space must be equal to the target feature space $x_S = x_T$. Before instance transfer,

Table 14
Sensors used in transfer learning-based human activity recognition articles.

Off-body sensor	Related paper links
Vision based	Alexiou et al. (2016), Antol et al. (2014), Ben-Ari et al. (2021), Bhattacharya et al. (2014), Brattoli et al. (2020), Cao et al. (2020), Careaga et al. (2019), Chang et al. (2020), Chen, Wang, et al. (2019), Coskun et al. (2021), de Souza et al. (2017), Du, Farrahi, and Niranjan (2019), Dwivedi et al. (2019), Feng and Duarte (2019), Feuz and Cook (2017), Gan, Lin, et al. (2016), Gan et al. (2015), Gan, Yang, et al. (2016), Gao et al. (2019), Haddad et al. (2021), Hong et al. (2015), Hu et al. (2019), Jain et al. (2015), Ji, Liu, Pang, and Li (2020), Ji, Liu, Pang, Ouyang, and Li (2020), Ji et al. (2018), Karpathy et al. (2014), Kay et al. (2017), Khan and Roy (2018), Kodirov et al. (2015), Liu et al. (2018), Liu, Liu, et al. (2019), Liu, Mei, et al. (2015), Liu et al. (2014), Liu, Su, et al. (2016), Luo et al. (2017), Mallya and Lazebnik (2016), Mandal et al. (2019), Mettes and Snoek (2017), Mishra et al. (2020, 2018), Morales and Roggen (2016), Narui et al. (2020), Pauly et al. (2021), Perrett et al. (2021), Prabono et al. (2020, 2021), Qi et al. (2020), Rey and Lukowicz (2017), Roitberg et al. (2017, 2018), Rokni and Ghasemzadeh (2017), Shen et al. (2018), Soleimani and Nazerfard (2021), Sun et al. (2021, 2015), Sztylet et al. (2017), Thaker and Krishnakumar (2017), Tran et al. (2015), Vondrick et al. (2016), Wang, Chang, et al. (2018), Wang and Chen (2017a, 2017b, 2020), Wang, Jiang, et al. (2018), Wang, Zheng, et al. (2018), Wen et al. (2016), Xu et al. (2015), Xu, Hospedales, and Gong (2016, 2017), Xu et al. (2020, 2018), Xu, Zhu, et al. (2016), Yamada et al. (2014), Yang et al. (2021), Ying et al. (2015), Zhang, Han, et al. (2017), Zhang, Li, and Reh (2017), Zhang et al. (2015, 2013, 2020, 2021), Zhou and Mu (2020), Zhu et al. (2018), Zhu and Newsam (2017), Zhu and Shao (2014a), Zhu et al. (2021), Zhu and Yang (2018), Zou et al. (2020, 2018)
Kinect	Cabrera et al. (2017), Cabrera and Wachs (2017), Du et al. (2020), Fu et al. (2019), Guo et al. (2018), Jasani and Mazagonwalla (2019), Jiang et al. (2015), Karn and Jiang (2016), Ke et al. (2017), Kviatkovsky et al. (2014), Lang et al. (2018), Li et al. (2021, 2018), Lin et al. (2015), Liu et al. (2014), Liu, Xu, et al. (2015), Ma et al. (2020), Memmesheimer et al. (2020), Pfister, Charles, and Zisserman (2014), Pikramenos et al. (2020), Rahmani and Mian (2015), Rodriguez et al. (2017), Sabater et al. (2021), Wan et al. (2015), Yang et al. (2012), Zhang, Li, and Ogunbona (2017), Zhang, Zhang, et al. (2017)
IR	Akbari and Jafari (2019), Feuz and Cook (2015), Li et al. (2020, 2018), Rokni and Ghasemzadeh (2017, 2018), Sanabria and Ye (2020), Ye (2018)
Door	de Souza et al. (2017), Feuz and Cook (2015)
Switch	Bhattacharya et al. (2014), Chang et al. (2020), Chen, Wang, et al. (2019), Du, Farrahi, and Niranjan (2019), Feng and Duarte (2019), Feuz and Cook (2017), Khan and Roy (2018), Morales and Roggen (2016), Prabono et al. (2020, 2021), Rey and Lukowicz (2017), Rokni and Ghasemzadeh (2017), Sanabria and Ye (2020), Soleimani and Nazerfard (2021), Wang, Zheng, et al. (2018), Wen et al. (2016)
Wi-Fi	Fetter and Gross (2014), Hong et al. (2015), Jiang et al. (2018), Shi, Zhang, Xu, and Cheng (2020), Shi, Zhang, Xu, Cheng, and Pearce (2020), Shu et al. (2018), Wang, Chen, et al. (2018)
Ultrasonic	Chen, Liu, et al. (2019), Du et al. (2018, 2020), Du, Jin, et al. (2019), Fu et al. (2021), He et al. (2019), Jiang et al. (2018), Lang et al. (2019), Li et al. (2020), Park et al. (2016), Seyfioglu et al. (2018), Shrestha et al. (2019)
Radio frequency	Laput et al. (2018), Papagiannaki et al. (2019), Sztylet et al. (2017), Yao et al. (2016)
Pressure	Feuz and Cook (2015), Sanabria and Ye (2020), Ye (2018)
Temperature	Feuz and Cook (2015), Papagiannaki et al. (2019), Wen and Zhong (2015)

we may bring the source and target domain feature into a common representation.

5.2.2. Feature representation transfer

In this TL, the distance between the underlying feature spaces of source and target domain data is tried to minimize through the following three mapping techniques:

1. Mapping source feature spaces into target feature spaces
2. Mapping target feature spaces into source feature spaces
3. Mapping both into a common feature space

To implement this learning technique, one should become familiar with two basic data operations, i.e., dimension reduction and data mapping operation. It helps project the source and target feature space into a common latent feature space.

5.2.3. Parameter transfer

The name itself suggests that this approach concentrates on learning the shared parameters between source and target tasks. The prior distribution of the hyper-parameters is also taken into account while implementing parameter TL. Many architectures use the Gaussian model to fetch prior distribution knowledge of source and target domain data. We assign equal weights to source and target domain parameters in multi-task learning as it encourages parallel learning of source and target data. However, in parameter transfer learning, we assign a larger weight to the target domain loss function of the target domain as compared to the source domain because we are interested in boosting the target domain by utilizing the parameter-related knowledge in the

source domain. It is also observed that the source weight W_s and target weight W_t is sometimes represented through the following expressions, i.e. $W_s = W_0 + V_s$ and $W_t = W_0 + V_t$. Where W_0 is the common weight term for both source and target. V_s and V_t stand for the domain-specific term.

5.2.4. Relational knowledge transfer

Rational knowledge transfer comes into action when the data is not independent and identically distributed (iid) instead of having multiple relationships. Network and graph are used to draw the relation between source and target task. In relational knowledge transfer, the domain-task relationship in the source dataset is transferred to the target dataset.

Among all of the above four TL techniques as shown in Tables 16–26, feature-based and parameter-based TL techniques are more frequently used.

5.3. Data and environment centric TL

When we gathered all the diversity while working on the activity recognition, the importance of transfer learning became valuable in terms of cost and effort, and accuracy. We represent three classification formats relevant to diversity in data (source data and target data) and environment.

5.3.1. TL based on cross-data transfer

Intra-dataset transfer and Inter-dataset transfer are the two standard transfer learning variants in activity recognition while taking data and

Table 15

Performance parameter used in transfer learning-based human activity recognition articles.

Performance parameter	Related paper links
Accuracy	Akbari and Jafari (2019), Al-Naser et al. (2018), Alam and Roy (2017), Antol et al. (2014), Ben-Ari et al. (2021), Brattoli et al. (2020), Cabrera et al. (2017), Cabrera and Wachs (2017), Cao et al. (2020), Careaga et al. (2019), Chavarriaga et al. (2013), Chen, Liu, et al. (2019), Chen, Wang, et al. (2019), Coskun et al. (2021), de Souza et al. (2017), Deng et al. (2020, 2014), Diethe et al. (2016), Ding et al. (2018), Du et al. (2018, 2020), Du, Jin, et al. (2019), Dwivedi et al. (2019), Fallahzadeh and Ghasemzadeh (2017), Feng and Duarte (2019), Fetter and Gross (2014), Feuz and Cook (2015, 2017), Fu et al. (2021, 2019), Gao et al. (2019), Guo et al. (2018), Haddad et al. (2021), Hong et al. (2015), Hu et al. (2019), Ijjina and Mohan (2014), Jain et al. (2015), Jänicke et al. (2016), Jasani and Mazagonwalla (2019), Ji, Liu, Pang, and Li (2020), Ji, Liu, Pang, Ouyang, and Li (2020), Ji et al. (2018), Jiang et al. (2018, 2015), Kalouris et al. (2019), Karpathy et al. (2014), Kay et al. (2017), Ke et al. (2017), Khan et al. (2018), Koldirov et al. (2015), Kviatkovsky et al. (2014), Lang et al. (2018), Laput et al. (2018), Li et al. (2020, 2021, 2018), Liang and Thomaz (2019), Liu et al. (2018), Liu, Liu, et al. (2019), Liu, Mei, et al. (2015), Liu et al. (2014), Liu, Su, et al. (2016), Liu, Wu, and Jia (2015), Liu, Xu, et al. (2015), Luo et al. (2017), Ma et al. (2020), Mallya and Lazebnik (2016), Mandal et al. (2019), Mazankiewicz et al. (2020), Memmesheimer et al. (2020), Mishra et al. (2020, 2018), Narui et al. (2020), Oh et al. (2021), Papagiannaki et al. (2019), Park et al. (2016), Pauly et al. (2021), Perrett et al. (2021), Prabono et al. (2020, 2021), Qi et al. (2020), Qin et al. (2019), Rahmani and Mian (2015), Rodriguez et al. (2017), Roitberg et al. (2017, 2018), Rokni and Ghasemzadeh (2017, 2018), Rokni et al. (2018), Saeedi et al. (2016, 2018), Sanabria and Ye (2020), Seyfioglu et al. (2018), Shi, Zhang, Xu, and Cheng (2020), Shi, Zhang, Xu, Cheng, and Pearce (2020), Shrestha et al. (2019), Shu et al. (2018), Soleimani and Nazerfard (2021), Sun et al. (2021), Thaker and Krishnakumar (2017), Tran et al. (2015), Vondrick et al. (2016), Wang and Chen (2017a, 2017b), Wang, Chen, et al. (2018), Wang, Jiang, et al. (2018), Wang, Zheng, et al. (2018), Wen and Zhong (2015), Xu, Hospedales, and Gong (2016, 2017), Xu et al. (2020, 2018), Xu, Zhu, et al. (2016), Yamada et al. (2014), Yang et al. (2021, 2012), Yao et al. (2016), Ye (2018), Ying et al. (2015), Zhang, Han, et al. (2017), Zhang, Li, and Ogunbona (2017), Zhang, Li, and Reh (2017), Zhang et al. (2015, 2013), Zhang, Zhang, et al. (2017), Zhang et al. (2020, 2021), Zhou and Mu (2020), Zhu et al. (2018), Zhu and Newsam (2017), Zhu and Shao (2014a), Zhu et al. (2021), Zhu and Yang (2018), Zou et al. (2020, 2018)
Precision	Al-Naser et al. (2018), Chang et al. (2020), Gan, Yang, et al. (2016), Hossain, Khan, and Roy (2017), Liu, Mei, et al. (2015), Pikramenos et al. (2020), Rokni and Ghasemzadeh (2017), Sabater et al. (2021), Saeed et al. (2019), Sun et al. (2015), Szttyler et al. (2017), Wang, Chang, et al. (2018), Wang and Chen (2020), Xu, Hospedales, and Gong (2016)
Recall	Al-Naser et al. (2018), Chang et al. (2020), Feuz and Cook (2015), Gan, Yang, et al. (2016), Hossain et al. (2017), Pfister et al. (2014), Pikramenos et al. (2020), Rokni and Ghasemzadeh (2017), Sabater et al. (2021), Saeed et al. (2019), Szttyler et al. (2017), Wang and Chen (2020)
F_1 score	Bhattacharya et al. (2014), Chang et al. (2020), Du, Farrahi, and Niranjani (2019), Feng and Duarte (2019), Gan, Yang, et al. (2016), He et al. (2019), Hossain et al. (2017), Khan and Roy (2018), Liang and Thomaz (2019), Morales and Roggen (2016), Pikramenos et al. (2020), Rey and Lukowicz (2017), Sabater et al. (2021), Saeed et al. (2019), Sanabria et al. (2021), Szttyler et al. (2017), Wan et al. (2015), Wang and Chen (2020), Wen et al. (2016)
MAP/MLD	Gan, Lin, et al. (2016), Gan et al. (2015), Mettes and Snoek (2017), Shen et al. (2018), Wang and Chen (2020), Xu et al. (2015) (Karn & Jiang, 2016; Lin et al., 2015)

Table 16

Review of existing works in HAR based on inductive transfer learning (2014–2021).

Reference	Architecture	Mode	Application	Dataset
Karpathy et al. (2014)	CNN	Feature	Activity classification	UCF-101, Sports-1M(S)
Bhattacharya et al. (2014)	NB+DT+NN	Feature	Activity recognition	OPPORTUNITY, challenge dataset2, Synthetic(S)
Ijjina and Mohan (2014)	CNN	Feature	Activity recognition	CMU Mocap – UCF-101
Kviatkovsky et al. (2014)	PCA+NN	Feature	Action recognition	Cambridge-Gestures, chaLearn, UCF-101
Liu et al. (2014)	BOW+SVM	Feature	Action recognition	KTH, TJU, MSR 3-D (2), MV-TJU
Sun et al. (2015)	CNN+LSTM	Parameter	Action recognition	FGA-240, THUMOS 2014/ ImageNet(S)
Liu, Mei, et al. (2015)	CNN	Feature	Thumbnails recognition	AMT/ Clickture(S)
Wen and Zhong (2015)	NB+HMM	Feature	Activity discovering	CASAS, Synthetic(S)
Jiang et al. (2015)	DTW+K-means	Feature	Gesture recognition	ChaLearn Gesture Dataset
Diethe et al. (2015)	Bayes	Instance	Activity in smart home	Smart home dataset
Lin et al. (2015)	MoSIFT+NN	Feature	Gesture recognition	ChaLearn Gesture Dataset
Liu, Xu, et al. (2015)	BoW + SVM	Feature	Action recognition	KTH, MV-TJU
Tran et al. (2015)	3D CNN+SVM	parameter	Action recognition	Sports-1M, UCF-101/I380K(S), ImageNet(S)
Hong et al. (2015)	SVM+BN	Feature	Activity recognition	Synthetic
Wen et al. (2016)	HMM+CRF	Instance	Action recognition	OPPORTUNITY
Park et al. (2016)	DCNN	Parameter	Activity recognition	Micro-Doppler dataset, ImageNet(S)

environment as a reference criteria. Intra-dataset transfer learning can be a reference to either cross-view or cross-action or cross-subjects

type of transfer learning. Inter-dataset transfer learning is also named cross-dataset transfer learning. Our paper summarizes all these four

Table 17
Review of existing works in HAR based on inductive transfer learning (2014–2021).

Link	Architecture	Mode	Application	Dataset
Diethé et al. (2016)	Bayes	Instance	Activity recognition	Synthetic
Xu, Zhu, et al. (2016)	SVM	Feature	Action recognition	UCF 50, UCF Sports, HMDB 51, Olympic Sports(S)
Vondrick et al. (2016)	GAN	Parameter	Action classification	Flickr
Wan et al. (2015)	BoVW+TS+NN	Feature	Activity recognition	MSR Daily Activity 3D, CAD-60
Karn and Jiang (2016)	K-NN	Feature	Gesture recognition	ChaLearn
Mallya and Lazebnik (2016)	Faster R-CNN	Instance	Interaction recognition	HICO, MPII Human Pose /ImageNet(S)
Liu, Su, et al. (2016)	Clustering	Feature	Action recognition	Hollywood2, YouTube, UCF Sports,UCF50, HMDB51, UCF101, KTH, TJU, MV-TJU, IXMAS
Rokni and Ghasemzadeh (2017)	SVM	Instance	Activity recognition	IRH(DV), OPPORTUNITY, SDA, IRH(SV)(S)
Zhu and Newsam (2017)	3D CNN	Parameter	Action detection	THUMOS 2014, MSR Actions II, MPII cooking, Sports1M(S)
de Souza et al. (2017)	Cool-TSN	Parameter	Action detection	UCF101, HMDB51, PHAV(S), MOCAP(S)
Rodriguez et al. (2017)	SHMM	Parameter	Activity recognition	Weizmann, IXMAS, KTH/HMDB51,OlympicSport,Virat 2.0
Cabrera and Wachs (2017)	HMM/SVM/CRF/DTW	Feature	Gesture recognition	ChaLearn2013, MSCR-12
Ke et al. (2017)	VGG-16	Feature	Action recognition	NTU RGB+D , SBU kinect-interaction, CMU

Table 18
Review of existing works in HAR based on inductive transfer learning (2014–2021) contd.

Link	Architecture	Mode	Application	Dataset
Rodriguez et al. (2017)	UBM+HMM	Parameter	Activity recognition	Weizmann, KTH, IXMAS, ChaLearn/HMDB51, Olympic Sprts, Virat 2.0(S)
Thaker and Krishnakumar (2017)	ResNet+LSTM	Parameter	Action recognition	Kinetics
Cabrera et al. (2017)	SVM/HMM	Instance	Gesture recognition	MSRC-12
Kay et al. (2017)	CNN+LSTM	Knowledge	Action recognition	UCF-101, HMDB-51/ Kinetics, ImageNet(S)
Zhang, Zhang, et al. (2017)	BoMW+NN	Parameter	Gesture recognition	RGB-D action, MSR gesture 3D, NTU RGB+D, MSRC-12, ChaLearn
Zhang, Zhang, et al. (2017)	BoMW+SVM	Feature	Gesture recognition	ChaLearn gesture, RGB MSR-gesture 3D, NTU+D, MSRC-12
Ji et al. (2018)	k-means	Relational-	Action early rec.	MSR Action Pairs, SYSU 3DHOI
Ye (2018)	NB/RF/SVM	Feature	Activity recognition	House-A, House-B, House-C
Du et al. (2018)	CNN(ResNet)	Parameter	Activity classification	Synthetic(T)/ImageNet (S)
Seyfioglu et al. (2018)	VGG/ResNet/DivNet	Parameter	Activitiy recognition	7-class, 11-class, ImageNet(S)
Rokni et al. (2018)	CNN	Feature	Activity recognition	WISDM, SDA(S)
Laput et al. (2018)	CNN(VGG-16)	Parameter	Activity recognition	AudioSet, FreeSound, Sound-snap, Network Sound, BBC Sound library, YouTube-8M(S)
Khan and Roy (2018)	TCA/JDA+SAE	Feature	Activity recognition	OPPORTUNITY, WISDM-Actitracker, Daily and Sports

classes, either under inductive transfer learning or transductive transfer learning, as shown in Tables 28 and 29.

1. *Cross-view Transfer Learning*: In some cases, multiple cameras with multiple installation points or viewpoints have been used while recording an action. Feature representations of these views of the same action are not similar and require different dictionaries to express. It is time-consuming and computationally expensive to design these dictionaries corresponding to each view for the same action separately. Hence, recordings across multiple views of the same action should be encouraged to get identical sparse representation. Cross-view transfer learning brings these distinct features into a common representation by transferring the knowledge from the source view to the target view.
2. *Cross-subject Transfer Learning*: Cross-subject TL refers to the sifting of knowledge between activities of the same class performed either by the different people of different appearances/different outfits/different groups/different gender/different gait patterns or by the same person at a different timestamp. Cross-subject

transfer learning learns the efficient way to transfer the knowledge from source to target within a dataset in an actor variable condition.

3. *Cross-action Transfer Learning*: Sometimes, we apply transfer learning across different actions within a dataset where the view and actors are the same. These actions should be related to each other by some means. Learning these related features and transferring them across activities are the primary themes of cross-action activity transfer learning. However, it is tough to get a good accuracy when cross-action transfer learning is applied between highly incongruent activities.
4. *Cross-dataset Transfer Learning*: Generally, cross-view, cross-subject, and cross-action transfer learning is confined to one dataset. To the best of our knowledge, we can interpret cross-dataset transfer learning as a combination of either cross-view and cross-subject or cross-view and cross-action. Here, the useful discriminative information is transferred from the source dataset to the target dataset. Regularization is frequently used in cross-dataset TL, which is a technique for projecting discriminative

Table 19

Review of existing works in HAR based on inductive transfer learning (2014–2021) contd.

Link	Architecture	Mode	Application	Dataset
Zhu and Yang (2018)	CMN+ResNet	Parameter	Action classification	Kinetics/ImageNet(S)
Xu et al. (2018)	TSN	Parameter	Activity recognition	UCF101/ Kinetics(S)
Zou et al. (2018)	GoogleNet+LSTM	Parameter	Action recognition	UCF11, UCF50, HMDB51
Guo et al. (2018)	NGM	Feature	Action recognition	CAD-120, PiGraphs
Du, Farrahi, and Niranjan (2019)	MLP, CL	Feature	Activity recognition	OPPORTUNITY, Skoda data(S)
Du, Jin, et al. (2019)	VGG/ResNet	Parameter	Activity classification	Micro-Doppler(T)/ImageNet(S)
He et al. (2019)	MRA-Net(CNN)	Feature	Activity recognition	micro-Doppler dataset
Lang et al. (2019)	JMI-CNN	Feature	Motion Classification	Synthetic
Shrestha et al. (2019)	AlexNet/GoogleNet	Parameter	Motion recognition	micro-Doppler, ImageNet(S)
Dwivedi et al. (2019)	GAN	Parameter	Action recognition	UCF101, HMDB51, Olympic-Sports, Sports1M(S)
Feng and Duarte (2019)	LSTM	Parameter	Activity recognition	OPPORTUNITY, PAMAP2
Papagiannaki et al. (2019)	SVM,CNN	Parameter	Activity recognition	FrailSafe
Fu et al. (2019)	ResNet-50	Parameter	Activity recognition	UnrealAction, DeepDrive, MiniKinetics(t)
Careaga et al. (2019)	AlexNet+LSTM	Parameter	Action recognition	Kinetics-600
Hu et al. (2019)	ResNet-18	Parameter	Action recognition	Kinetics-CMN
Deng et al. (2020)	CNN+WPN	Instance	Action recognition	PAMAP2, Skoda, UCI-HAR
Zou et al. (2020)	GoogleNet	Parameter	Action recognition	UCF11, HMDB51, ImageNet
Shi, Zhang, Xu, and Cheng (2020)	CNN+LSTM	Parameter	Activity recognition	Synthetic
Qi et al. (2020)	ResNet-50	Parameter	Activity recognition	Kinetics, Charades
Memmesheimer et al. (2020)	Deep residual CNN	Parameter	Action recognition	NTU RGB+D 120, UTD-MHAD

Table 20

Review of existing works in HAR based on inductive transfer learning (2014–2021) contd.

Link	Architecture	Mode	Application	Dataset
Zhang et al. (2020)	ARN (3D-CNN)	Feature	Action recognition	HMDB51, UCF101, miniMIT, Kinetics
Ji, Liu, Pang, Ouyang, and Li (2020)	ResNet-18, GCN	Parameter	Activity recognition	HICO-NN, TUHOI-NN, HICO-NF, TUHOI-NF
Cao et al. (2020)	DTW + ResNet-50	Parameter	Activity classification	Kinetics, Something-Something-V2
Zhou and Mu (2020)	ResNet+Attention+LSTM	Parameter	Activity classification	ActivityNet, Kinetics-400,
Xu et al. (2020)	3D ConvNet	Parameter	Activity detection	(ActivityNet1.2, ActivityNet1.3, THUMOS14
Ji, Liu, Pang, and Li (2020)	ResNet-18	Parameter	Interaction recognition	HICO-FS, TUHOI-FS/ ImageNet(S)
Ma et al. (2020)	LSTM	Feature	Gesture recognition	DHGD Hand Gesture, MSRA Hand Gesture
Oh et al. (2021)	DNN	Instance	Activity recognition	Synthetic, UCI(S)
Pauly et al. (2021)	3D-CNN	Parameter	Action recognition	MIME, :UCF101
Li et al. (2021)	Inception-ResNet/SVM	Parameter	Gesture recognition	Chalearn Isolated Gesture
Fu et al. (2021)	CNN+Bi-LSTM	Feature	Excercise recognition	Synthetic
Haddad et al. (2021)	K-means clustering	Feature	Action recognition	KTH,Weizmann
Coskun et al. (2021)	Inception-ResNet	Parameter	Action Recognition	EPIC, EGTEA,GUN71/ImageNet(S)
Zhu et al. (2021)	TSN+ResNet	Parameter	Action recognition	Kinetics-400, UCF101 , Sth-Sth-100 , HMDB-51
Yang et al. (2021)	I3D	Parameter	Action localization	AVA, CF101-24/ Kinetics (S)
Zhang et al. (2021)	ITA Net	Feature	Activity classification	Something-Something-V2, Kinetics
Sabater et al. (2021)	TCN	Parameter	Action recognition	NTU RGB+D 120, therapy dataset
Ben-Ari et al. (2021)	3D-CNN	Parameter	Action recognition	Kinetics-400, ActivityNet2/Sports1M(S)
Perrett et al. (2021)	ResNet-50	Parameter	Action recognition	Kinetics, Something-Something V2 (SSv2), HMDB51, UCF101

features of both source and target datasets into a common latent space.

5.3.2. TL based on task and environment conditionality

We can segregate learning techniques in the activity domain into four categories while relating to task and environment conditionality. These can be entitled Same Task Same Environment (STSE), Same Task Different Environment (STDE), Different Task Same Environment (DTSE), and Different Task Different Environment (DTDE). The sensor's position (height, distance, angle, and coverage area) from the actor in the human activity and the background settings of that activity are considered while analyzing the environment setting.

1. **STSE:** STSE (Deng et al., 2014; Haddad et al., 2021; He et al., 2019; Liu, Xu, et al., 2015; Morales & Roggen, 2016; Papagian-naki et al., 2019; Saeedi et al., 2018; Zhang, Li, & Ogunbona, 2017; Zhang, Li, & Rehg, 2017) class is straightforward learning

which requires some basic transfer learning rules to put on. Here, the task and environment settings of the involved activities are the same.

2. **STDE:** In STDE (Bagaveyev & Cook, 2014; Chen, Wang, et al., 2019; de Souza et al., 2017; Du, Farrahi, & Niranjan, 2019; Feng & Duarte, 2019; Fu et al., 2019; Jasani & Mazagonwalla, 2019; Khan et al., 2018; Kodirov et al., 2015; Liu, Mei, et al., 2015; Mallya & Lazebnik, 2016; Mishra et al., 2018; Oh et al., 2021; Rodriguez et al., 2017; Rokni & Ghasemzadeh, 2017; Sanabria et al., 2021; Wang & Chen, 2017b; Xu, Hospedales, & Gong, 2016)class, the tag itself defines that performed task is similar, but the environment settings may differ in many aspects. Gait, cognitive health, and tremor measurement in dementia patients at the hospital and at home can be taken as concrete illustrations of STDE transfer class.
3. **DTSE:** Two different groups of activities under similar environ-mental settings can be studied beneath the DTSE (Cao et al.,

Table 21
Review of existing works in HAR based on transductive transfer learning (2014–2021).

Link	Architecture	Mode	Application	Dataset
Zhu and Shao (2014a)	k-SVD	Feature	Action recognition	IXMAS
Deng et al. (2014)	TransRKELM	Feature	Activity recognition	UCI ML Repository
Zhu and Shao (2014b)	K-SVD	Parameter	Action recognition	UCF YouTube(T)/HMDB51(S)
Yamada et al. (2014)	Kernel regression+TGP	Instance	Activity recognition	Poser Dataset, HUMANEVA-I
Deng et al. (2014)	TransRKELM	Feature	Activity recognition	activation recognition (UCI)
Fetter and Gross (2014)	Naïve Bayes	Instance	Sensor activity rec.	IM, XML
Antol et al. (2014)	GRNN	Parameter	Interaction recognition	INTERACT, PARSE
Pfister et al. (2014)	DTW + SVM	Feature	Gesture recognition	BSL-TV sign language, BSL dictionary, ChaLearn gesture
Ying et al. (2015)	MSDTL	Feature	Activity recognition	Walk8, DaSA, HAR
Feuz and Cook (2015)	Naive Bayes	Feature	Activity recognition	Newsgroups
Gan et al. (2015)	Regression/SVM	Feature	Activity recognition	UCF101
Jain et al. (2015)	PCA + SVM	Feature	Activity classification	UCF101, THUMOS14, HMDB51, UCF Sports/YFCC100M(S)
Zhang et al. (2015)	RankSVM	Feature	Action recognition	KTH, UIUC, HOLLYWOOD2
Xu et al. (2015)	NN	Feature	Action recognition	HMDB51, UCF101 /Google News(S)
Kodirov et al. (2015)	CNN+NN	Feature	Action recognition	UCF-101/Thumos(S)
Saeedi et al. (2016)	BFADM, CADM, MADM	Feature	Activity recognition	UCI HAR
Jänicke et al. (2016)	Probabilistic CMM	Feature	Activity recognition	Synthetic, AR
Morales and Roggen (2016)	CNN + LSTM	Feature	Mobile activity rec.	OPPORTUNITY, Skoda Mini Checkpoint

Table 22
Review of existing works in HAR based on transductive transfer learning (2014–2021) contd.

Link	Architecture	Mode	Application	Dataset
Gan, Lin, et al. (2016)	Linear SVM	Feature	Video Activity rec.	TRECVID, UCF101, FCVID, GoogleSports1M, ActivityNet
Alexiou et al. (2016)	SESA, IAP.	Feature	Action recognition	UCF-101, HMDB-51
Xu, Hospedales, and Gong (2016)	RR+NN	Instance	Action recognition	HMDB51, UCF101, Olympic Sports
Xu, Hospedales, and Gong (2017)	Normalizes NN	Instance	Action recognition	HMDB51, UCF101, Olympic Sports, CCV, TRECVID/ImageNet(S)
Feuz and Cook (2017)	Regression/k-NN/SVM/DT	Feature	Activity recognition	Toy dataset, OPPORTUNITY, CASAS PUCK, CASAS Parkinson
Alam and Roy (2017)	Hierarchical SVM	Relational knowledge	Activity recognition	RCC, TU Darmstadt
Zhang, Li, and Ogunbona (2017)	JGSA + RBF	Feature	Action recognition	MSRAAction3DExt, G3D, UTD-MHAD, MAD
Fallahzadeh and Ghasemzadeh (2017)	K-means/Naïve bayes	Relational knowledge	Activity recognition	UCI Daily and Sports Activities
Rey and Lukowicz (2017)	Probabilistic	Relational knowledge	Activity recognition	OPPORTUNITY, bike
Szttyler et al. (2017)	Random forest	Feature	Activity recognition	Synthetic
Zhang, Han, et al. (2017)	IVA	Feature	Action recognition	UCF101, HMDB51, MVY /Stanford40, ASD, EAD, SIF(S)
Wang and Chen (2017a)	CNN	Parameter	Action recognition	UCF101, HMDB51
Zhang, Li, and Reh (2017)	R-CNN(AlexNet)	Parameter	Action recognition	GTEA, GTEA gaze+/ImageNet(S)
Roitberg et al. (2017)	BNN + I3D	Parameter	Action recognition	UCF-101, HMDB-51
Mettes and Snoek (2017)	Faster R-CNN	Parameter	Action recognition	UCF Sports, UCF 101, J-HMDB, Holly-wood2Tubes/YFCC100M,MS-COCO(S)
Wang and Chen (2017b)	GoogleNet/VGG	Parameter	Visual recognition	AwA, CUB-200, UCF101, HMDB51

2020; He et al., 2019; Hu et al., 2019; Lin et al., 2015; Ma et al., 2020; Mallya & Lazebnik, 2016; Pauly et al., 2021; Shi, Zhang, Xu, & Cheng, 2020; Wen et al., 2016; Zhang et al., 2020) transfer class.

4. **DTDE**: If the task and the environment of a performed activity change in transfer learning, it will be counted under the DTDE category. A person's character pattern analysis in the home and again in an office environment may come under DTDE transfer class.

TL can be closely compared with some machine transfer techniques like self-taught learning, multi-task learning, one-shot learning, few-shot learning, zero-shot learning, active learning, Stratified transfer learning, domain adaptation, and covariate shift. These models are meant to transfer the knowledge between two different data types with a slight variation in either data arrangement or management. Some of the are already discussed in Section 5.1.1, and the rest are described below.

Zero-shot learning (ZSL). ZSL (Al-Naser et al., 2018; Alexiou et al., 2016; Antol et al., 2014; Brattoli et al., 2020; Gan, Lin, et al., 2016; Gan et al., 2015; Jain et al., 2015; Jasani & Mazagonwalla, 2019; Kodirov et al., 2015; Liu, Liu, et al., 2019; Mandal et al., 2019; Mettes & Snoek, 2017; Mishra et al., 2020, 2018; Roitberg et al., 2017, 2018; Shen et al., 2018; Sun et al., 2021; Wang & Chen, 2017a, 2017b, 2020; Xu et al., 2015; Xu, Hospedales, & Gong, 2016, 2017; Yamada et al., 2014; Zhang, Li, & Reh, 2017; Zhang et al., 2015; Zhu et al., 2018) is a transductive transfer learning phenomenon between disjoint training and evaluation classes. The labeled training examples of seen activities leverage unlabeled examples of unseen activities for classification by sharing their inherent semantic knowledge. In ZSL, the key features in source data are analyzed and categorized as vectors with descriptions that enable the algorithm to classify unseen activities. Many efficient algorithms on ZSL have been rolling since 2008 (Okoli & Schabram, 2010). As mentioned in Table VIII and Table IX, we try to include some of them based on their significance towards activity recognition from 2014 to 2021.

Table 23

Review of existing works in HAR based on transductive transfer learning (2014–2021) contd.

Link	Architecture	Mode	Application	Dataset
Khan et al. (2018)	HD-CNN	Instance	Activity recognition	Synthetic smart phone/watch data
Wang, Chen, et al. (2018)	MMD+STL+SVM /Random forest	Feature	Activity recognition	OPPORTUNITY, PAMAP2, UCI Daily, Sports Activities
Wang, Zheng, et al. (2018)	CNN+LSTM	Feature	Activity recognition	OPPORTUNITY, UCI Daily, Sports Activities
Shu et al. (2018)	DIRT-T+VADA	Feature	Activity recognition	Wi-Fi Activity Recognition
Ding et al. (2018)	MMD+CNN	Feature	Activity recognition	UCI daily and sports, USC-HAD
Lang et al. (2018)	CNN+k-NN	Feature	Motion classification	Synthetic, micro-Doppler/MOCAP(S)
Saeedi et al. (2018)	K-means+RF	Feature	Activity recognition	UCI Daily and Sports Activities UCI Daily, Sports Activities
Jiang et al. (2018)	CNN	Feature	Activity recognition	CSI, ultrasound, mmWave, Visible Light
Liu et al. (2018)	3D-CNN	Parameter	Action recognition	UCF101, HMDB51/Sport-1M, Kinetics(S)
Mishra et al. (2018)	3D-CNN	Parameter	Action recognition	UCF101, HMDB51, Olympic
Al-Naser et al. (2018)	SVM	Feature	Action recognition	Synthetic
Shen et al. (2018)	R-CNN(VGG)	Parameter	Interaction recognition	HICODET
Roitberg et al. (2018)	I3D(CNN)	Feature	Action recognition	HMDB-51/Kinetics(S)
Zhu et al. (2018)	ResNet-200+TSN	Parameter	Action recognition	UCF101, HMDB51/Activity Net, ImageNet(S)
Akbari and Jafari (2019)	CNN	Feature	Activity recognition	HHAR, PAMAP, MoST
Chen, Liu, et al. (2019)	MMD/GAN	Feature	Activity recognition	μ -DS

Table 24

Review of existing works in HAR based on transductive transfer learning (2014–2021) contd.

Link	Architecture	Mode	Application	Dataset
Qin et al. (2019)	ASTTL + KNN	Relational- knowledge	Activity recognition	UCI Daily and Sports Activities, UCI-HAR, USC-HAD, PAMAP2
Liang and Thomaz (2019)	PCA + DCNN	Parameter	Activity recognition	Freesound, Urbansound, Audio- PairBank, Audioset, SoundNet
Kalouris et al. (2019)	CNN	Parameter	Activity recognition	FrailSafe(T)/ PAMAP2, UCI HAR(S)
Papagiannaki et al. (2019)	CNN	Parameter	Activity recognition	FrailSafe
Chen, Wang, et al. (2019)	MMD+NN+ STL-SAT(SVM)	Feature	Activity recognition	OPPORTUNITY, PAMAP2, UCI DSADS, MHEALTH
Liu, Liu, et al. (2019)	3D-CNN	Parameter	Action recognition	ActivityNet, Surv5H
Gao et al. (2019)	TS-GCN	Instance	Action recognition	Olympic Sports, HMDB51, UCF101
Wang and Chen (2020)	3D-CNN	Instance	Action recognition	Breakfast, Charades
Mandal et al. (2019)	3D-CNN	Parameter	Action recognition	Olympic Sports, HMDB51, UCF101/Sports-1M(S)
Jasani and Mazagonwalla (2019)	ST-GCN	Relational- knowledge	Skeletal action rec.	NTU-RGB+D House-C, Aruba, Twor
Sanabria and Ye (2020)	VAE	Feature	Activity recognition	House-A, House-B,
Prabono et al. (2020)	CNN(HydraNet)	Parameter	Activity recognition	OPPORTUNITY, UCI daily, sports activities
Narui et al. (2020)	1-D CNN	Feature	Activity recognition	WiFi Activity
Li et al. (2020)	CNN	Feature	Activity recognition	micro-doppler

Table 25

Review of existing works in HAR based on transductive transfer learning (2014–2021) contd.

Link	Architecture	Mode	Application	Dataset
Du et al. (2020)	CNN	Feature	Activity classification	CMU/Kinect-Based MOCAP
Mazankiewicz et al. (2020)	DA-BN + CNN	Feature	Activity recognition	WISDM
Chang et al. (2020)	PCA+MMD+DCNN	Feature	Activity recognition	RealWorld, OPPORTUNITY, HHAR, PAMAP2
Pikramenos et al. (2020)	CNN and RNN	Feature	Action recognition	PKU-MMD
Brattoli et al. (2020)	3D-CNN	Parameter	Action classification	UCF101, HMDB51, Activity- Net/Kinetics(S)
Mishra et al. (2020)	VAE, Bi-GAN	Parameter	Action recognition	UCF101, HMDB51, Olympic, PAMAP UCI Daily and Sports Activities
Prabono et al. (2021)	User defined CNN	Feature	Activity recognition	OPPORTUNITY, UCI Daily and Sports
Sun et al. (2021)	GAN, GCN	Relational- knowledge	Activity recognition	UCF101, HMDB51
Soleimani and Nazerfard (2021)	SA-GAN	Instance	Activity recognition	OPPORTUNITY

One-shot learning (OSL) & few-shot learning (FSL). OSL (Cabrera et al., 2017; Cabrera & Wachs, 2017; Fu et al., 2019; Haddad et al., 2021; Ijjina & Mohan, 2014; Ji et al., 2018; Jiang et al., 2015; Karn & Jiang, 2016; Kviatkovsky et al., 2014; Li et al., 2021; Ma et al., 2020; Memmesheimer et al., 2020; Pauly et al., 2021; Rodriguez et al., 2017, 2017; Sabater et al., 2021; Shi, Zhang, Xu, & Cheng, 2020; Wan et al.,

2015; Zhang, Zhang, et al., 2017; Zou et al., 2020, 2018) and FSL (Ben-Ari et al., 2021; Cao et al., 2020; Coskun et al., 2021; Deng et al., 2020; Fu et al., 2021; Ji, Liu, Pang, & Li, 2020; Perrett et al., 2021; Qi et al., 2020; Thaker & Krishnakumar, 2017; Xu et al., 2020, 2018; Yang et al., 2021; Zhang et al., 2020, 2021; Zhou & Mu, 2020; Zhu et al., 2021; Zhu & Yang, 2018) can be categorized under the inductive transfer learning technique, requiring one and a few labeled examples of unseen

Table 26

Review of existing works in HAR based on unsupervised transfer learning (2014–2021).

Link	Architecture	Mode	Purpose	Dataset	Cross Transfer
Chavarriaga et al. (2013)	SVM	Feature	Action recognnition	IXMAS, N-UCLA	Cross view
Rahmani and Mian (2015)	NKTM(SVM)	Feature	Action recognnition	IXMAS, N-UCLA	Cross view
Sanabria et al. (2021)	Bi-GAN+SVM	Feature	Action recognnition	IXMAS, N-UCLA	Cross action/dataset
Rokni and Ghasemzadeh (2018)	DT,kNN, NB	Feature	Activity Recognition	Synthetic (UCI repo)	Cross view
Li et al. (2018)	3D-CNN+RNN	Feature	Skeletal activity rec.	NTU RGB+D	Cross subject/view
Wang, Jiang, et al. (2018)	3D-CNN+RNN	Feature	Activity Recognition	SOMETHING-SOMETHING	Cross action/dataset
Saeed et al. (2019)	TPN(CNN)	Feature	Activity Recognition	HHAR, UniMiB, UCI HAR, MobiAct, WISDM, MotionSense	Cross subject

Table 27

Related transfer learning articles based on label information settings.

Type based on label availability	Related paper link
RS learning	Ben-Ari et al. (2021) , Cabrera et al. (2017) , Cabrera and Wachs (2017) , Cao et al. (2020) , Careaga et al. (2019) , Coskun et al. (2021) , de Souza et al. (2017) , Deng et al. (2020) , Diethe et al. (2015, 2016) , Du, Farrahi, and Niranjana (2019) , Du et al. (2018) , Du, Jin, et al. (2019) , Feng and Duarte (2019) , Fu et al. (2021, 2019) , Gan, Yang, et al. (2016) , Guo et al. (2018) , Haddad et al. (2021) , He et al. (2019) , Hong et al. (2015) , Hu et al. (2019) , Ijjina and Mohan (2014) , Ji, Liu, Pang, and Li (2020) , Ji, Liu, Pang, Ouyang, and Li (2020) , Ji et al. (2018) , Jiang et al. (2015) , Karn and Jiang (2016) , Karpathy et al. (2014) , Kay et al. (2017) , Ke et al. (2017) , Khan and Roy (2018) , Kviatkovsky et al. (2014) , Lang et al. (2019) , Laput et al. (2018) , Li et al. (2021) , Lin et al. (2015) , Liu, Mei, et al. (2015) , Liu et al. (2014) , Liu, Su, et al. (2016) , Liu, Wu, and Jia (2015) , Liu, Xu, et al. (2015) , Ma et al. (2020) , Memmesheimer et al. (2020) , Oh et al. (2021) , Papagiannaki et al. (2019) , Park et al. (2016) , Paul, Bappy, and Roy-Chowdhury (2017) , Pauly et al. (2021) , Perrett et al. (2021) , Qi et al. (2020) , Rodriguez et al. (2017) , Rokni and Ghasemzadeh (2017) , Rokni et al. (2018) , Sabater et al. (2021) , Seyfioglu et al. (2018) , Shi, Zhang, Xu, and Cheng (2020) , Shrestha et al. (2019) , Sun et al. (2015) , Thaker and Krishnakumar (2017) , Tran et al. (2015) , Vondrick et al. (2016) , Wan et al. (2015) , Wang, Zheng, Huang, and Ding (2017) , Wen et al. (2016) , Wen and Zhong (2015) , Xu et al. (2020, 2018) , Xu, Zhu, et al. (2016) , Yang et al. (2021) , Ye (2018) , Zhang, Zhang, et al. (2017) , Zhang et al. (2020, 2021) , Zhou and Mu (2020) , Zhu and Newsam (2017) , Zhu et al. (2021) , Zhu and Yang (2018) , Zou et al. (2020, 2018)
RU learning	Bhattacharya et al. (2014) , Dwivedi et al. (2019) , Mallya and Lazebnik (2016)
US learning	Akbari and Jafari (2019) , Al-Naser et al. (2018) , Alam and Roy (2017) , Alexiou et al. (2016) , Antol et al. (2014) , Brattoli et al. (2020) , Chang et al. (2020) , Chen, Liu, et al. (2019) , Chen, Wang, et al. (2019) , Deng et al. (2014) , Ding et al. (2018) , Du et al. (2020) , Fallahzadeh and Ghasemzadeh (2017) , Fetter and Gross (2014) , Feuz and Cook (2015, 2017) , Gan, Lin, et al. (2016) , Gan et al. (2015) , Gao et al. (2019) , Jänicke et al. (2016) , Jasani and Mazagonwalla (2019) , Jiang et al. (2018) , Kalouris et al. (2019) , Khan et al. (2018) , Kodirov et al. (2015) , Lang et al. (2018) , Li et al. (2020) , Liang and Thomaz (2019) , Liu et al. (2018) , Liu, Liu, et al. (2019) , Mandal et al. (2019) , Mazankiewicz et al. (2020) , Mettes and Snoek (2017) , Mishra et al. (2020, 2018) , Morales and Roggen (2016) , Pfister et al. (2014) , Pikramenos et al. (2020) , Prabono et al. (2020, 2021) , Qin et al. (2019) , Rey and Lukowicz (2017) , Roitberg et al. (2017, 2018) , Saeedi et al. (2016, 2018) , Sanabria and Ye (2020) , Sanabria et al. (2021) , Shen et al. (2018) , Shu et al. (2018) , Soleimani and Nazerfard (2021) , Sun et al. (2021) , Sztylet et al. (2017) , Wang and Chen (2017a, 2017b, 2020) , Wang, Chen, et al. (2018) , Wang, Zheng, et al. (2018) , Xu et al. (2015) , Xu, Hospedales, and Gong (2016, 2017) , Yamada et al. (2014) , Ye (2018) , Ying et al. (2015) , Zhang, Han, et al. (2017) , Zhang, Li, and Ogunbona (2017) , Zhang et al. (2015) , Zhu et al. (2018) , Zhu and Shao (2014a)
UU learning	Chen and Lin (2014) , Fei and Li (2020) , Han et al. (2018) , Lu et al. (2016) , Moriya and Shibata (2018) , Okoli and Schabram (2010) , Zhu and Shao (2014b)

Table 28

Related cross transfer articles under inductive platform.

Type of cross TL	Related paper link
Cross action inductive TL	Ben-Ari et al. (2021) , Cabrera et al. (2017) , Cabrera and Wachs (2017) , Cao et al. (2020) , Careaga et al. (2019) , Coskun et al. (2021) , Deng et al. (2020) , Diethe et al. (2016) , Du et al. (2018) , Fu et al. (2019) , Gan, Yang, et al. (2016) , Guo et al. (2018) , He et al. (2019) , Hu et al. (2019) , Ji, Liu, Pang, and Li (2020) , Ji, Liu, Pang, Ouyang, and Li (2020) , Ji et al. (2018) , Karn and Jiang (2016) , Kviatkovsky et al. (2014) , Lang et al. (2019) , Li et al. (2021) , Lin et al. (2015) , Ma et al. (2020) , Mallya and Lazebnik (2016) , Memmesheimer et al. (2020) , Pauly et al. (2021) , Perrett et al. (2021) , Qi et al. (2020) , Rodriguez et al. (2017) , Sabater et al. (2021) , Shi, Zhang, Xu, and Cheng (2020) , Thaker and Krishnakumar (2017) , Vondrick et al. (2016) , Wen et al. (2016) , Xu et al. (2020) , Yang et al. (2021) , Zhang, Zhang, et al. (2017) , Zhang et al. (2020, 2021) , Zhou and Mu (2020) , Zhu et al. (2021) , Zhu and Yang (2018) , Zou et al. (2020, 2018)
Cross view inductive TL	Diethe et al. (2015) , Fu et al. (2021, 2019) , Ke et al. (2017) , Liu, Su, et al. (2016) , Rokni and Ghasemzadeh (2017)
Cross dataset inductive TL	Bagaveyev and Cook (2014) , Bhattacharya et al. (2014) , Coskun et al. (2021) , de Souza et al. (2017) , Du, Farrahi, and Niranjana (2019) , Du, Jin, et al. (2019) , Dwivedi et al. (2019) , Feng and Duarte (2019) , Fu et al. (2019) , Garcia-Ceja and Brena (2015) , Jiang et al. (2015) , Karpathy et al. (2014) , Kay et al. (2017) , Khan and Roy (2018) , Laput et al. (2018) , Liu and Ferrari (2017) , Liu, Mei, et al. (2015) , Liu et al. (2014) , Liu, Su, et al. (2016) , Mallya and Lazebnik (2016) , Oh et al. (2021) , Park et al. (2016) , Rodriguez et al. (2017) , Rokni et al. (2018) , Seyfioglu et al. (2018) , Shrestha et al. (2019) , Sun et al. (2015) , Tran et al. (2015) , Wan et al. (2015) , Wen and Zhong (2015) , Xu et al. (2018) , Xu, Zhu, et al. (2016) , Yang, Ma, Nie, Chang, and Hauptmann (2015) , Zhang, Zhang, et al. (2017) , Zhu and Newsam (2017) , Zhu and Yang (2018) , Zou et al. (2020)
Cross subject inductive TL	Diethe et al. (2015) , Haddad et al. (2021) , He et al. (2019) , Ijjina and Mohan (2014) , Ke et al. (2017) , Liu, Xu, et al. (2015) , Papagiannaki et al. (2019) , Zhang, Zhang, et al. (2017)

Table 29
Related cross transfer articles under transductive platform.

Type of cross transfer	Related link
Cross action transductive	Akbari and Jafari (2019), Al-Naser et al. (2018), Alam and Roy (2017), Alexiou et al. (2016), Antol et al. (2014), Brattoli et al. (2020), Chen, Liu, et al. (2019), Chen, Wang, et al. (2019), Ding et al. (2018), Fetter and Gross (2014), Feuz and Cook (2017), Gan, Lin, et al. (2016), Gan et al. (2015), Gao et al. (2019), Jasani and Mazagonwalla (2019), Jiang et al. (2018), Khan et al. (2018), Kodirov et al. (2015), Liang and Thomaz (2019), Liu, Liu, et al. (2019), Mandal et al. (2019), Mazankiewicz et al. (2020), Mettes and Snoek (2017), Mishra et al. (2020, 2018), Narui et al. (2020), Prabono et al. (2020, 2021), Roitberg et al. (2017, 2018), Sanabria and Ye (2020), Sanabria et al. (2021), Shen et al. (2018), Shu et al. (2018), Sun et al. (2021), Wang and Chen (2017a, 2017b, 2020), Wang, Chen, et al. (2018), Wang, Zheng, et al. (2018), Xu et al. (2015), Xu, Hospedales, and Gong (2016, 2017), Ying et al. (2015), Zhang et al. (2015), Zhu et al. (2018)
Cross view transductive	Chen, Wang, et al. (2019), Gan et al. (2015), Khan et al. (2018), Saeedi et al. (2016), Shu et al. (2018), Zhu and Shao (2014a)
Cross dataset transductive	Akbari and Jafari (2019), Brattoli et al. (2020), Chang et al. (2020), Chen, Wang, et al. (2019), Feuz and Cook (2017), Jain et al. (2015), Liang and Thomaz (2019), Liu et al. (2018), Liu, Liu, et al. (2019), Mettes and Snoek (2017), Pfister et al. (2014), Prabono et al. (2020), Qin et al. (2019), Rey and Lukowicz (2017), Roitberg et al. (2017, 2018), Sanabria and Ye (2020), Sanabria et al. (2021), Wang and Chen (2017a), Wang, Chen, et al. (2018), Wang, Zheng, et al. (2018), Xu et al. (2015), Ying et al. (2015), Zhang, Han, et al. (2017), Zhang, Li, and Ogunbona (2017)
Cross subject transductive	Deng et al. (2014), Ding et al. (2018), Fallahzadeh and Ghasemzadeh (2017), Kalouris et al. (2019), Morales and Roggen (2016), Papagiannaki et al. (2019), Pikramenos et al. (2020), Saeedi et al. (2016, 2018), Soleimani and Nazerfard (2021), Szttyler et al. (2017), Yamada et al. (2014), Zhang, Li, and Ogunbona (2017), Zhang, Li, and Rehgi (2017)

Table 30
Performance recognized in various transfer learning enhanced articles 2014–2021.

References	Transfer mode	Target dataset	Accuracy(%)	Application
Karpathy et al. (2014)	Inductive	UCF101	63.3	H-O/Group level AR
Bhattacharya et al. (2014)	Inductive	Synthetic	79.0	Atomic and interaction AR
Liu et al. (2014)	Inductive	KTH	96.7	Atomic AR
	Inductive	TJU	97.6	Activity of daily living
	Inductive	Daily Activity 3D	90.6	Activity of daily living
Gan et al. (2015)	Transductive	UCF101	84.64	H-O/Group level AR
Jain et al. (2015)	Transductive	EPIC	62.2	Kitchen AR
Sun et al. (2015)	Inductive	Thumus 2014	73.6	Atomic/H-O/Group level AR
Wen and Zhong (2015)	Inductive	CASAS	88.63	Activity of daily living
Zhang et al. (2015)	Transductive	Hollywood	61.73	H-O/H-H interaction
	Transductive	KTH	95.98	Atomic AR
Xu, Hospedales, and Gong (2017)	Transductive	HMDB51	56.4	H-O/H-H level AR
	Transductive	UCF101	82.0	H-O/Group level AR
	Transductive	Olympic Sports	93.4	Sports Action/Interaction
Liu, Xu, et al. (2015)	Inductive	MV-TJU	89.6	Multi-view daily activity
Tran et al. (2015)	Inductive	UCF101	52.8	H-O/Group level AR
Wang and Chen (2017a)	Transductive	UCF101	50.67	H-O/Group level AR
	Transductive	HMDB51	31.51	H-O/H-H level AR
Roitberg et al. (2017)	Transductive	UCF101	88.66	H-O/Group level AR
	Transductive	HMDB51	97.52	H-O/H-H level AR
Wang and Chen (2017a)	Inductive	UCF101	40.12	H-O/Group level AR
	Inductive	HMDB51	25.82	H-O/H-H level AR
Wen et al. (2016)	Inductive	Opportunity	81.3	Activity of daily living
Park et al. (2016)	Inductive	Micro doppler	80.3	Gesture level AR
Hong et al. (2015)	Inductive	Synthetic	83.4	Atomic and interaction AR
Wang and Chen (2017b)	Transductive	UCF101	66.9	H-O/Group level AR
	Transductive	HMDB51	22.3	H-O/H-H level AR
Wang, Chen, et al. (2018)	Transductive	Opportunity	43.78	Activity of daily living
	Transductive	PAMAP	37.83	Atomic/H-O AR
Xu, Zhu, et al. (2016)	Inductive	HMDB51	82	H-O/H-H level AR
	Inductive	UCF50	82	Group sports AR
	Inductive	Olympic Sports	87	Sports Action/Interaction

activities. In both cases, the representation vectors are constructed by separately taking the critical features from the labeled seen and unseen activities. After doing so, the similarity algorithm is modeled by taking these optimum distances. The similarity algorithm will later decide the fate of the target domain activity, i.e., from which category that target domain activity belongs.

Active transfer learning (ATL). The annotation effort while building an effective supervised learning model can be minimized through the help of active learning. Transfer learning helps to exploit these annotated data, extract valuable features from them and utilize them to learn

different but related unseen data. These two techniques can be forged together to form the ATL (Diethe et al., 2015; Fu et al., 2021; Mallia & Lazebnik, 2016; Oh et al., 2021) model, which can effectively reduce the labeling effort and enhance the overall performance. This model is called ATL.

Stratified transfer learning (STL). STL (Bhattacharya et al., 2014) is another variant of transductive transfer learning that can exploit intra-affinity between activity classes to execute intra-class knowledge transfer. It helps to improve the classification accuracy for cross-domain

Table 31
Performance recognized in various transfer learning enhanced articles 2014–2021(Cont').

References	Transfer mode	Target dataset	Accuracy(%)	Application
Wan et al. (2015)	Inductive	CAD50	85.42	Atomic/H-O level AR
	Inductive	MSR3D	95.7	Atomic/H-O level AR
Liu, Su, et al. (2016)	Inductive	TJU	96.9	Activity of daily living
	Inductive	KTH	94.3	Atomic AR
	Inductive	IXMAS	94.7	Atomic multi-view AR
Liu et al. (2018)	Transductive	UCF101	92.5	H-O/Group level AR
	Transductive	HMDB51	62.8	H-O/H-H level AR
Mishra et al. (2018)	Transductive	UCF101	78.68	H-O/Group level AR
	Transductive	HMDB51	52.58	H-O/H-H level AR
	Transductive	Olympics sports	83.81	Sports Action/Interaction
de Souza et al. (2017)	Inductive	UCF101	94.	H-O/Group level AR
	Inductive	HMDB51	69.5	H-O/H-H level AR
Rodriguez et al. (2017)	Inductive	Weizman	81.5	Atomic AR
	Inductive	KTH	74	Atomic AR
	Inductive	IXMAS	47.4	Atomic multi-view AR
Ke et al. (2017)	Inductive	NTU-RGB	84.8	Atomic/H-O/H-H Level
	Inductive	SBU Kinect	93.57	H-H interaction
	Inductive	CMU	88.3	Kitchen AR
Thaker and Krishnakumar (2017)	Inductive	Kinetics	78	H-O/H-H level AR
Roitberg et al. (2018)	Transductive	HMDB51+Kinetics	34.77	H-O/H-H level AR
	Transductive	UCF101	49.9	H-O/Group level AR
	Transductive	HMDB51	58.2	H-O/Group level AR
Zhu and Yang (2018)	Inductive	Kinetics	78.9	H-O/H-H level AR
Xu et al. (2018)	Inductive	UCF101	86.23	H-O/Group level AR
Dwivedi et al. (2019)	Inductive	HMDB51	54	H-O/Group level AR
	Inductive	Olympics Sports	86	Sports Action/Interaction
	Inductive	UCF101	80	H-O/Group level AR
Careaga et al. (2019)	Inductive	Kinetics	84.2	H-O/H-H level AR
Soleimani and Nazerfard (2021)	Transductive	Oppertunity	73	Activity of daily living
Qin et al. (2019)	Transductive	UCI-HAR	57.8	Activities of daily living
	Transductive	PAMAP	54.7	Atomic/H-O AR
Zou et al. (2020)	Inductive	UCF101	81.8	H-O/Group level AR
	Inductive	HMDB51	51	H-O/H-H level AR
Memmesheimer et al. (2020)	Inductive	NTU-RGBD	49.51	Atomic/H-O/H-H Level

activity recognition. STL performs a majority voting technique to obtain pseudo target labels before applying for Intra-class transfer.

6. Performance and computational complexity in TL enhanced HAR

6.1. Performance analysis

In this chapter, we summarize and discuss transfer learning enhanced HAR articles from 2014 to 2021 in terms of performance (accuracy), transfer mode, target dataset, and mode of application. Most preferably, we include articles from inductive and transductive backgrounds. The techniques used in these articles are previously tabulated in Tables 16–25. Tables 30–32 distinctively tagged transfer learning-based HAR articles with regard to these parameters.

We can observe that the use of inductive transfer learning is frequent over transductive learning, where the accuracy does matter. Due to the presence of target domain samples, the inductive models are more efficiently trained than the transductive models. As a result, inductive models yield more accurate performances than the other, which can clearly be recognized in the tables mentioned earlier. The model in Zhu et al. (2021) adopts the inductive learning technique on the UCF-101 dataset and produces an accuracy of 87.1%. Whereas the same relative model in Brattoli et al. (2020) only can yield an accuracy of 48 % on the very same UCF-101 dataset. The performance of a model highly depends on the complexity of a dataset. As observed in Brattoli et al. (2020), the accuracies for UCF-101 and HMDB-51 are 48 % and 32.7 %, respectively. But for both the observation, the mode of transfer and model architecture remain unaltered. The activity and the background in the HMDB-51 dataset are more complex than UCF-101. The same inference can be pulled out throughout these tables.

6.2. Computational complexity

We have also investigated the computational complexity of various machine learning models that recognize several human activities under the transfer learning domain. During this extensive investigation, we came across many relative parameters that unveiled “How economical a model could be in terms of computational cost and speed?”. The computational cost and speed of an algorithm are estimated by constantly keeping an eye on the models’ performance. These relative parameters are total recognition time, time/epoch, total learnable parameters, average run time, required memory, order of calculation, and Frame Per Second (FPS). As shown in Table 33, three different comparison criteria, i.e., time/epoch, total parameter, and convolutional parameter, are picked up in Seyfioglu et al. (2018) to express relative computational complexity between CNN, VGGNet, ResNet, and DivNet-15 techniques. It is measured on the Microdoppler dataset using K80 GPU, having 24GB RAM. As shown in Table 34, Wan et al. (2015) segregates four different descriptors in terms of average run time and measured accuracies on the MSR Activity 3D dataset. Jiang et al. (2015) expresses the computational complexity of four ML techniques on the ChaLearn Gesture dataset in the form of total recognized time, shown in Table 34. Kay et al. (2017), Thaker and Krishnakumar (2017) compare distinct deep learning models by counting the total learnable parameters, as represented in Table 35. The computational complexity of four descriptors is analyzed by calculating the proposed model’s runtime, FPS, and speed on the UCF-101 dataset, shown in Table 36 (Tran et al., 2015). Wang and Chen (2020) has compared computational complexity between visual and semantic models and between Charades and Breakfast datasets. They count total learnable parameters for model-based complexity. For dataset-based comparison, time/epoch, total time, and memory count have been considered, as shown in Table 37. Liu et al. (2018) calculate computational complexity by observing speed (in terms of FPS) and accuracy on the UCF-101 dataset. Zhang, Han, et al. (2017) compare the average computational time with respect

Table 32
Performance recognized in various transfer learning enhanced articles 2014–2021(Cont').

References	Transfer mode	Target dataset	Accuracy(%)	Application
Zhang et al. (2020)	Inductive	HMDB51	60.56	H-O/H-H level AR
	Inductive	UCF101	82.97	H-O/Group level AR
Zhou and Mu (2020)	Inductive	ActivityNet	90.76	H-O level AR
	Inductive	Kinetics400	88.96	H-O/H-H level AR
Kalouris et al. (2019)	Transductive	Frail safe	84.89	Health-care AR
Chen, Wang, et al. (2019)	Transductive	Oppertunity	46.8	Activity of daily living
	Transductive	PAMAP	49.3	Atomic/H-O AR
Xu et al. (2020)	Inductive	ActivityNet 1.2	34.2	H-O level AR
	Inductive	ActivityNet 1.3	34.6	H-O level AR
	Inductive	Thumos14	28.1	Atomic/H-O/Group level AR
Haddad et al. (2021)	Inductive	Weizmann	89.4	Atomic AR
	Inductive	KTH	73	Atomic AR
Gao et al. (2019)	Transductive	Olympics sports	59.9	Sports Action/Interaction
	Transductive	HMDB51	31	H-O/H-H level AR
Gao et al. (2019)	Transductive	UCF101	41.6	H-O/Group level AR
Mandal et al. (2019)	Transductive	Olympics sports	66.2	Sports Action/Interaction
	Transductive	HMDB51	36.1	H-O/H-H level AR
	Transductive	UCF101	49.4	H-O/Group level AR
Zhu et al. (2021)	Inductive	Kinetics 100	87.1	H-O/H-H level AR
	Inductive	HMDB51	75.8	H-O/H-H level AR
	Inductive	UCF101	95.2	H-O/Group level AR
Jasani and Mazagonwalla (2019)	Transductive	NTU-RGBD	75.16	Atomic/H-O/H-H Level
Brattoli et al. (2020)	Transductive	UCF101	48	H-O/Group level AR
	Transductive	HMDB51	32.7	H-O/H-H level AR
Brattoli et al. (2020)	Transductive	ActivityNet	26.6	H-O level AR
Zhang et al. (2021)	Inductive	Kinetics100	84.3	H-O/H-H level AR
Ben-Ari et al. (2021)	Inductive	Activity 1.2	83.12	H-O level AR
Mishra et al. (2020)	Transductive	Olympics sports	49.80	Sports Action/Interaction
	Transductive	UCF101	23.86	H-O/Group level AR
	Transductive	HMBD51	18.73	H-O/H-H level AR
Sun et al. (2021)	Transductive	UCF101	37.6	H-O/Group level AR
	Transductive	HMBD51	36.4	H-O/H-H level AR
Perrett et al. (2021)	Inductive	Kinetics	85.9	H-O/H-H level AR
	Inductive	SSV2	64.6	H-O level AR
	Inductive	HMBD51	75.6	H-O/H-H level AR
	Inductive	UCF101	96.1	H-O/Group level AR

Table 33
Observed computational complexity in Seyfioglu et al. (2018).

Approach	Time/Epoch (s)	Total parameter	Convolution parameter
CNN	0.4	124 525	28 064
VGGNet	9	152,00,000	14,700,000
ResNet	20	14,700,000	23,581,712
DivNet	1	4,814,000	1,614,091

to accuracy to find complexity and effectiveness between different transfer learning descriptors. The computational complexity in Jiang et al. (2015), Seyfioglu et al. (2018), Thaker and Krishnakumar (2017), Tran et al. (2015), Wan et al. (2015) and Kay et al. (2017) are estimated under the inductive TL platform, and the complexity in Liu et al. (2018), Wang and Chen (2020) and Zhang, Han, et al. (2017) are evaluated under the transductive platform.

7. Challenges and future research directions

7.1. Challenges

If we take a deep dive into this subject, we will encounter many challenges that need to be fixed in the future to make the transfer learning-based HAR context more robust and generalized. These problems may be hypothetically considered the future direction that needs greater attention among the research fraternity. We split these challenges into five tiers, which are described below:

7.1.1. Dataset oriented challenges

- How to choose a source dataset that is best suited for a given target data?

- How to handle data imbalance problems while using transfer learning techniques in HAR?
- How to efficiently handle the large-scale source domain data?

7.1.2. Pre-processing oriented challenges

- How to extract valuable knowledge from noisy source domain data?
- What are the rules for evaluating and exploiting the more competent feature that will be more suited for target domain data?
- How can video-level pre-processing be effectively implemented across long sequence video data?

7.1.3. Learning oriented challenges

- What are the most compelling feature selection criteria and techniques for a successful knowledge transfer?
- How to flourish the powerful transfer learning techniques like zero-shot transfer learning, and unsupervised transfer learning?
- The hidden negotiable relationships between human activity recognition datasets can potentially enhance the performance of the transfer learning-based HAR model. How to create a platform to enhance some undervalued learning ideas like relational knowledge transfer?

7.1.4. Model oriented challenges

- Some models are working fine for a particular kind of dataset but not so well for others. So, how to increase the model's flexibility that will perform optimally for every class of datasets?
- How to enhance the generalizability of the model irrespective of any transfer learning variant?
- How to take control of overfitting criterion and variance problems during model training?

Table 34

Observed computational complexity in [Wan et al. \(2015\)](#) using CPU: i7 4790 @3.6 GHz, 24 GB RAM (left); Observed computational complexity in [Jiang et al. \(2015\)](#) using DELL PC: Duo CPU E8400 (Right).

Descriptors	Average runtime	Accuracy(%)	Approach	Rec. Time (s)
MFSK	98 ms/f	41.2	DTW	2.6
HOG+HOF	18 ms/f	32.1	CHMM	6.89
MBH	18 ms/f	27.3	SCHMM	6.79
3D SMOsIFT	0.32 ms/f	36.7	SOFM	6.74

Table 35

Observed computational complexity in [Thaker and Krishnakumar \(2017\)](#) on Kinetics dataset (left); Observed computational complexity in [Kay et al. \(2017\)](#) (Right).

Approach	Total parameter	Approach	Parameter
AlexNet+LSTM	4 million	ConvNet+LSTM	29 M
LSTM	10 million	Two stream	48 M
VGG+LSTM	25 million	3D ConvNet	79 M

Table 36

Observed computational complexity in [Tran et al. \(2015\)](#).

Descriptors	Runtime	FPS	Slower speed
Improved Dense Trajectory (IDT) (CPU)	202.2	3.5	91.4
Brox's Optical Flow (CPU)	2513.9	0.3	1135.9
Brox's Optical Flow (GPU)	607.8	1.2	274.6
3D	2.2 h	313.9	1

Table 37

Observed computational complexity in [Wang and Chen \(2020\)](#) using GTX 1080 Ti GPU.

Dataset	Time/Epoch ((sec/epoch)	Total time	Memory (GB)
Charades	40	13 min	3.5
Breakfast	300	1 hr	10.5

Table 38

Future scope related to transfer learning-enhanced HAR.

Area for future investigation	Description	Possible solutions
Energy and resource optimization	Real-time sensing and model adaption needs Continuous energy flow and resources consumption	• Reduction in irrelevance features • Adaptive frequency domain analysis • Sampling frequency optimization • Adopt local and global processing unit systems • Data augmentation in space and time
Insufficient training data collection	Unavailability of diverse data in terms of scene, subject (age and body shapes, height), background noises and illumination	• Advance transfer learning technology • Improved model learning algorithm • Fake but resemble data creation • Time-scale invariant feature description • Semi-supervised learning technology • Evidence boosting methods

7.1.5. Algorithm oriented challenges

- How to make the entire architecture robust for online and real-time human activity recognition?
- Most of the papers are based on single-mode input data. More than one mode could enhance the performance of the model architecture. How to model the transfer learning architecture so as to perform well on multi-modal source data?
- It is incredibly challenging to realize the hierarchical framework of complex high-level activities because they encompass deeper semantic and context information. How do we develop an architecture that exploits these high-level activities' co-relation to yield enhanced performance?

7.2. Future directions

It is incredibly challenging to realize the hierarchical framework of complex high-level activities because they encompass deeper semantic and context information. So, we need to develop an architecture that

exploits these high-level attributes to yield enhanced performance. Many efforts have been put to make the entire architecture robust for real-time recognition. Certainly, this goal is not yet achieved but seems not far from reality. Our paper investigates concerning areas that come in the way of building a recognition model online. We have tried to spotlight those scopes along with future recommendations. The future scopes and their probable solutions are listed in the below [Tables 38–41](#).

8. Conclusion

The underneath concept of this comprehensive survey using state-of-the-art transfer learning methods is to ease the hardship and effort behind data collection, data extinction, and data labeling along with accuracy enhancement in the activity recognition domain. This paper focuses on the HAR in context-aware applications and empathizes its diversity with transfer learning functionality. We intuitively explore the statistical research interest in transfer learning-based HAR through

Table 39

Future scope related to transfer learning-enhanced HAR.

Area for future investigation	Description	Possible solutions
Activity transition and overlapping	Challenging to detect and recognize activity when abnormal transition or simultaneous occurrence of two activities takes place	• Confuse the classifier while performing multiple tasks at the same time • Proficient spatiotemporal feature learning • Effective smoothing • Scene transition modeling • Co-ordinate or axial position shifting
Placement of the sensor	• Variation in the raw input data due to sensor orientation and position • Variation in wearable sensor location due to different behavioral pattern during same activity • Different background variation (variation in camera location)	• Abolishing certain component by feature modeling
Sensor selection and deployment	Selecting an inappropriate sensor for a particular activity led to decline in recognition performance	• Efficient sensor fusion technique • Led to compromise in performance, when sensor is placed in a wrong location • Follow one-to-one, one-to-many, many-to-one, many-to-many sensor deployment scenarios • Study the predefined ontology for sensor selection • Adopt data level, feature level, and decision level fusion techniques
Multimodel recognition	Improper synchronization and fusing of data collected from multiple sensor devices yields inferior recognition accuracy	• Investigate ambient and wearable sensor fusion mechanism
Low level recognition	Negatively affect by interference, illumination, noises and occlusion. So require low level recognition system	• Low level to high level feature extraction • Consider local to global features • Apply techniques like nearest point detection and bag of words classification

Table 40

Future scope related to transfer learning-enhanced HAR.

Area for future investigation	Description	Possible solutions
Contextual information	Challenging to understand the scene without contextual information	• Recognize low level motion features • Track and recognize interacting and non interacting object • Keep knowledge about state-of-the-art like facial expression, distance features, object interaction
Real-time recognition	Recognize algorithm may loose its significance if it fails to detect and identify activity in real time	• On-board processing (local processing) leads to reduce computation cost in terms of storage and transfer of data • Proper parameter tuning • Efficient initialization techniques • Optimal compression technique • Optimize the hyper-parameters
Computational complexity	Computational complexity of the model has a greater impact of recognition time, speed and accuracy	• Dimensionality reduction • Effective regularization and normalization techniques • Design efficient filter for noise reduction useful feature selection • Cost sensitive learning strategy
Class imbalance	Hard to collect examples of rare occurring classes that influence performance of learning algorithm	• Weighted learning schemes • Over sampling and active learning • Synthetic instance generation

this article by following the Okali and Schabram methodology. We try to investigate the research priority related to type, mode, domain, and label availability of transfer learning. The whole-length depiction, investigation, and high points of this paper help the researcher to achieve in-depth knowledge in activity recognition using transfer learning techniques. We have presented a comprehensive sketch of sensor-based HAR using transfer learning. The key points of this paper are to analyze the underlying limitations faced while recognizing these activities and their transfer learning-oriented solutions. Most of the discussion focuses on “how transfer learning advocates its presence while

dealing with activity-related problems like outdated, recollection, and labeling effort behind training dataset?”. The objective of this survey is to communicate the limitations and their optimal solution related to TL-based HAR.

Apart from transfer learning and their extensive applications in activity recognition, other various orientations still lie down on the floor to investigate and discover for subsequent research such as detection, tracking, design, and classification. This survey does not comprehensively address the learning algorithms used in transfer learning-based HAR. In the future, we will extend this work by featuring different

Table 41
Future scope related to transfer learning-enhanced HAR.

Area for future investigation	Description	Possible solutions
Data labeling strategy	Manual labeling strategies are expensive and faulty	• Take the help of crowd-sourcing platform • EX. Amazon Mechanical Turk (AMT) annotating strategy • Keep knowledge about state-of-the-art like facial expression, distance features, object interaction • Adopting zero-shot and semi-supervised learning strategies • Pseudo learning reduces the overfitting and improves speed
Hardware limitation	• Real-time recognition become harder due to computational constraints • Compromise with input data or in model architecture	• Explore the techniques like Cropping, compression, sub-shot, thresholding, key-frame and key-point extraction • Trade-off between computational complexity and performance • Design informative sensing technology and feature-extracting descriptors
Negative transfer	Sometimes target domain performance deteriorates after coupling with the source data	• Finding the origin of negative transfer or factor influencing negative transfer • Finding selective features that minimize the domain divergence • Improve the source and target domain data quality • Adopt preventive approach like secure transfer mechanism, domain similarity algorithm, and distance transfer assessment
Dataset quality and diversity	Several improvements are required in terms of quality, diversity and availability within dataset for a competitive and interpretive HAR algorithms and training machine learning models	• Increasing the diversity of participants by including individuals of different ethnicities, ages, genders, heights, and weights • Capturing data from participants wearing a variety of clothing and accessories and recording multiple complete gait cycles for each participant • Capturing data in multiple real-world settings, including indoor and outdoor locations • Including RGB data in the dataset • Recording data from multiple angles, annotating it, and making it available to the public

classes of learning algorithms related to this domain. This inclusive survey is supposed to strengthen further research in activity recognition grassland.

CRedit authorship contribution statement

Abhisek Ray: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Formal analysis, Investigation. **Ma-heshkumar H. Kolekar:** Writing – review, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix. Transfer learning

Transfer learning is a technique of boosting performance measures during test/validation that is trained on a data set belonging to either a different feature space or distinct distribution. The feature space and distribution of the data change from one environment to another, from one device to another, and from one setting to another. Every time, We need to develop a fresh trained model from the grass-roots level to cope with this changing feature space and distribution. However, it

Table A.42
Terminology used in transfer learning.

Terminology	Notation
Source domain	D_S
Target domain	D_T
Source task	T_S
Target task	T_T
Source feature space	x_S
Target feature space	x_T
Source label space	y_S
Target label space	y_T
Source marginal probability distribution	$P_S(X)$
Target marginal probability distribution	$P_T(X)$
Source conditional probability distribution	$P_S(Y/X)$
Target conditional probability distribution	$P_T(Y/X)$

is a tedious, time-consuming, and expensive phenomenon to cultivate a whole new setup every time by collecting a new set of training data. Researchers intelligently utilize the previously learned knowledge in a different task or domain environment to reduce the effort behind extensive training data recollection. Unavailability of enough training data, compensation for outdated data, and relaxation for data collection and labeling efforts pilot us into a sphere of knowledge transfer. Transfer learning can be applied in classification, regression, clustering (Karpathy et al., 2014), and reinforcement learning tasks. The following section lists and depicts some key terms (also listed in Table A.42) and definitions, categories, and applications.

This section discusses some key terms like source, target, domain, and task with different notations. Here, we show how the relationship between these four helps to deliver the meaning of transfer learning.

Table A.43
Machine learning vs transfer learning.

Case	Sub-case	Learning
$D_S = D_T$ and $T_S = T_T$	$x_S = x_T$ & $P_S(X) = P_T(X)$ and $y_S = y_T$ & $P_S(Y/X) = P_T(Y/X)$	ML
$D_S \neq D_T$	$x_S \neq x_T$ or $P_S(X) \neq P_T(X)$	TL
$T_S \neq T_T$	$y_S \neq y_T$ or $P_S(Y/X) \neq P_T(Y/X)$	TL

Source and target. Transfer learning opens the gate to extracting valuable information from abundant data and utilizing them to exploit the sparse data. The source is that abundance database where we find and collect useful related information to learn our model. However, the target comprises the scarce data that need to be exploited by utilizing that learned information.

Domain. The complete set of knowledge that gives an idea about features and their distribution probability is called domain (D). In transfer learning, a feature space (χ) is a collection of features obtained from the source or target data, and the marginal probability of a particular feature among a set of features is known as the marginal probability distribution ($P(X)$). A domain that $D = \{ \chi, P(X) \}$, comprises of two factors:

1. Feature space χ ;
2. Marginal probability distribution $P(X)$, where $X = \{x_1, x_2, x_3, \dots, x_n\} \in \chi$.

In terms of source and target, we can define a *domain* as the source domain (D_S) and the target domain (D_T). Two different domains imply source and target datasets have either separate feature space ($\chi_S \neq \chi_T$); or separate marginal probability distribution ($P(\chi_S) \neq P(\chi_T)$).

Task. In a very similar manner, a task (T) is defined as the data aggregation of object label information and its probability of prediction at a given specific domain. Label space (γ) is nothing but the collection of class labels associated with a set of source or target class objects, and the probability distribution of a class object at a given specific domain feature space is called a conditional probability distribution ($P(Y/X)$). A task that $T = \{ \gamma, P(Y/X) \}$, comprises two factors:

1. Label space γ ;
2. Conditional probability distribution $P(Y/X)$, where $Y = \{y_1, y_2, y_3, \dots, y_n\} \in \gamma$ and $X = \{x_1, x_2, x_3, \dots, x_n\} \in \chi$. $P(Y/X)$ is to be learned by the pairs $\{x_i, y_i\}$.

In terms of source and target, we can also define a *task* as the source task (T_S) and the target task (T_T). Two different tasks imply source and target datasets have either separate label space ($\gamma_S \neq \gamma_T$); or separate marginal probability distribution ($P(Y_S/X_S) \neq P(Y_T/X_T)$).

Transfer learning is the concept of breaking free from the isolated learning paradigm and exploiting information obtained for one task to unravel correlated ones. Case of transfer learning arises only when $D_S = D_T$ and $T_S = T_T$ conditions do not hold good as shown in Table A.43. Transfer learning is a strategy for improving test/validation performance by training datasets from different feature spaces or distributions. The pivotal motivation behind the scene is to boost the objective predictive function in the target domain by taking the knowledge of the source domain (D_S), source learning task (T_S), target domain (D_T) and target learning task (T_T).

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